UNIT – I

Elastic & Plastic Behaviour – Dislocations

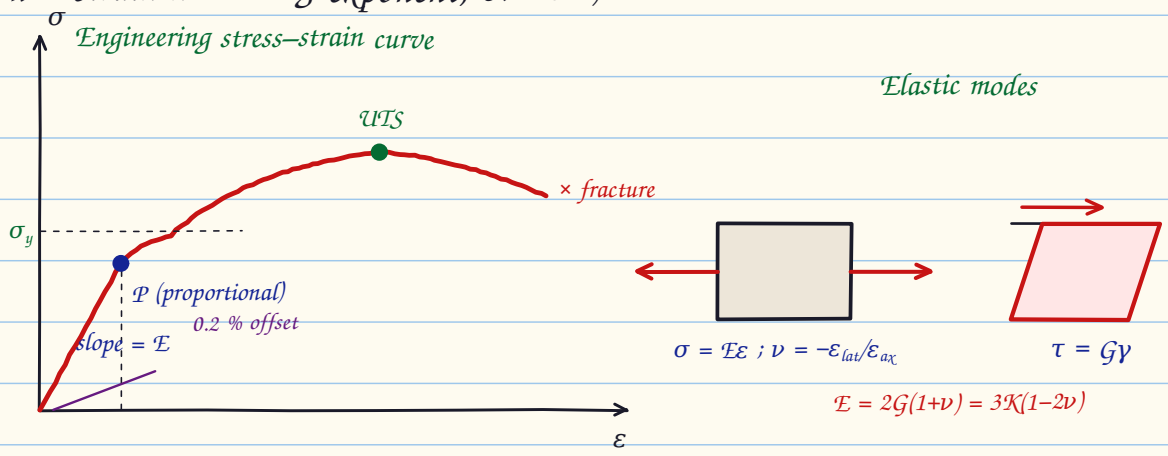
Subject : Mechanical Behaviour of Materials (23A03504c)

Syllabus: Elastic behaviour – Hooke's law ; plastic behaviour – dislocation theory, Burgers vectors, dislocation loops, dislocations in FCC, HCP & BCC lattices, stress fields & energies of dislocations, forces on and between dislocations, slip and twinning.

1. Introduction – Mechanical Behaviour

Mechanical behaviour = response of a material (deformation, yielding, fracture) to applied loads. Engineers need strength, stiffness, ductility, toughness, fatigue & creep resistance.

- **Engineering stress** $\sigma = P/A_0$; **engineering strain** $\varepsilon = \Delta l/l_0$. **True stress** $\sigma_T = P/A = \sigma(1+\varepsilon)$; **true strain** $\varepsilon_T = \ln(l/l_0) = \ln(1+\varepsilon)$ (valid up to necking, constant volume).
- Plastic flow after yield follows $\sigma_T = K\varepsilon_T^{\,n}$ (Hollomon : K = strength coefficient, n = strain-hardening exponent, 0.1–0.5).



- Regions of the stress–strain curve : elastic (reversible, Hooke), yield (onset of plasticity), strain hardening (up to UTS), necking and fracture. Yield strength is usually the **0.2 % offset** stress (line parallel to the elastic slope drawn from $\varepsilon = 0.002$).

- Ductility : % elongation = $(l_f - l_0)/l_0 \times 100$; % reduction of area = $(A_0 - A_f)/A_0 \times 100$. Toughness = area under the σ–ε curve ; resilience $U_r = \sigma_y^2/2E$.

2. Elastic Behaviour – Hooke's Law

Elastic deformation is reversible : the atoms are displaced from their equilibrium positions but return when the load is removed (bond stretching).

$$\sigma = E\,\varepsilon \;\; \tau = G\,\gamma \;\; p = -K(\Delta V/V) \text{ (Hooke's law)}$$

- **E** Young's modulus (tension), **G** shear (rigidity) modulus, **K** bulk modulus. **Poisson's ratio** $\nu = -\varepsilon_{lateral}/\varepsilon_{axial}$ (0.25–0.35 for metals, 0.5 for incompressible rubber, $\nu \le 0.5$).

$$E = 2G(1+\nu) = 3K(1-2\nu) \quad G = E/2(1+\nu) \quad K = E/3(1-2\nu)$$

- **Origin** : interatomic force $F(r)$ – at the equilibrium spacing r_0 the net force is zero ; for small displacement $F \approx S_0(r - r_0)$, where $S_0 = dF/dr$ (bond stiffness). $E \propto S_0/r_0$ ⇒ strong, tightly bonded materials (high T_m) have high E ; E drops slowly with temperature.
- E is a structure-insensitive property (not changed by grain size, cold work, heat treatment) but depends on crystal direction (anisotropy) : Cu $E_{[111]} = 191$ GPa, $E_{[100]} = 67$ GPa.

Material	E (GPa)	G (GPa)	ν	Remark
Steel	200–210	79	0.29	stiff
Cast iron	100–170	45	0.26	
Aluminium	70	26	0.33	1/3 of steel
Copper	120	44	0.34	
Titanium	110	42	0.34	
Glass / concrete	70 / 30	28 / 12	0.22 / 0.2	brittle

Generalised Hooke's law (isotropic, 3-D)

$$\varepsilon_x = (1/E)[\sigma_x - \nu(\sigma_y + \sigma_z)] \quad \gamma_{xy} = \tau_{xy}/G \text{ (and cyclic)}$$

- Dilatation $\Delta = \varepsilon_x + \varepsilon_y + \varepsilon_z = (1-2\nu)(\sigma_x + \sigma_y + \sigma_z)/E = p/K$. Anisotropic crystals need up to 21 independent constants (cubic : C_{11}, C_{12}, C_{44}) ; anisotropy

ratio $A = 2C_{44}/(C_{11} - C_{12})$.

- **Anelasticity** : time-dependent recovery of strain (internal friction) – damping in metals ; **elastic limit** may differ slightly from the proportional limit.

> Ex 1. $E = 200$ GPa, $\nu = 0.30$: $G = 200/2.6 =$ 76.9 GPa ; $K = 200/1.2 =$ 166.7 GPa. A steel rod loaded to $\sigma = 400$ MPa : $\varepsilon = 400/200\,000 = 0.002$; lateral strain $= -0.3(0.002) = -0.0006$. Resilience for $\sigma_y = 500$ MPa : $U_r = 500^2/(2 \times 200\,000) =$ 0.625 MJ/m³.

3. Plastic Behaviour – Why Dislocations?

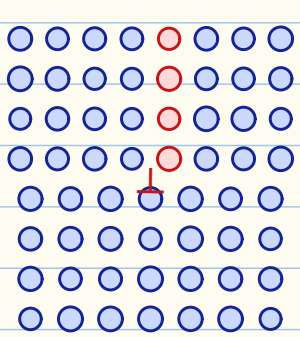
Plastic deformation = permanent shape change by slip (movement of atoms to new equilibrium positions) beyond the elastic limit, at constant volume.

- **Theoretical shear strength** (Frenkel, 1926) : if two perfect planes slide over each other as rigid blocks, $\tau = (G\,b/2\pi\,a)\sin(2\pi x/b) \Rightarrow \tau_{th} \approx G/2\pi \approx G/10 - G/30 \approx$ 3–8 GPa for metals.
- **Observed** yield strength of pure single crystals is only 0.1–10 MPa ($10^{-3} - 10^{-4}$ G) – 1000 times lower ⇒ slip does not occur by simultaneous shift of whole planes but by the motion of line defects, **dislocations** (Orowan, Polanyi, Taylor – 1934).
- Analogy : moving a carpet by a ripple ; caterpillar motion ; only a few bonds are broken at a time ⇒ low stress.

> Ex 2. Cu : $G = 48$ GPa ⇒ $\tau_{th} = 48/(2\pi) = 7.6$ GPa ; measured CRSS ≈ 0.6 MPa ⇒ ratio ≈ 10^4 – evidence for dislocations. Dislocations were first seen directly by TEM (Hirsch, 1956) ; density ρ = line length per unit volume (m^{-2}).

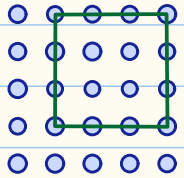
4. Dislocations – Types & Burgers Vector

Edge dislocation

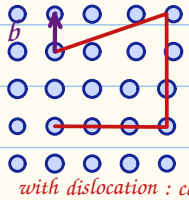


b ⊥ dislocation line ; ξ ⊥ b

Burgers circuit



perfect : closes



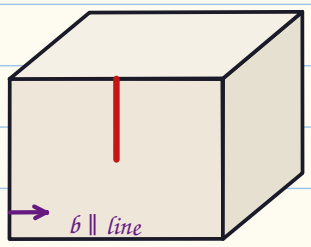
with dislocation : closure failure = b

Edge dislocation

- Formed by an extra half-plane of atoms inserted in a perfect lattice. The dislocation line ξ is the edge of the half-plane. Above the line the lattice is in compression, below in tension. Symbol ⊥ (positive, extra plane above) / ⊤ (negative).
- **Burgers vector b** : the closure failure of a circuit (Burgers circuit) which is closed in the perfect crystal – b is perpendicular to the line in an edge dislocation. Slip occurs on the plane containing b and ξ (a unique glide plane) ; edge can also **climb** (move out of the glide plane by vacancy diffusion – needs thermal activation).
- **Burgers circuit rule** : (FS/RH convention) circuit drawn clockwise looking along ξ ; b = vector from finish to start (F→S).

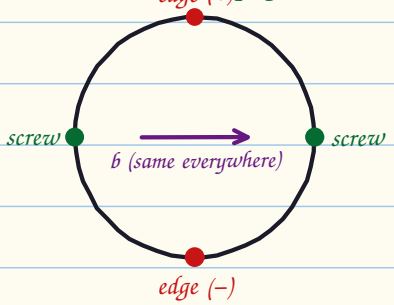
Screw dislocation

Screw dislocation

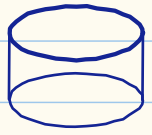


spiral ramp ; shear τ ∥ line

Dislocation loop (glide)



Prismatic loop



b ⊥ loop plane (climb)

vacancy / interstitial disc

- Crystal planes form a continuous spiral ramp (like a screw) around the line ; b is parallel to the line. Produced by shear applied to part of the crystal. Since b ∥ ξ, any plane containing the line is a glide plane ⇒ the screw can **cross-slip** (change

slip plane) – important in recovery and work hardening ; it does not climb.

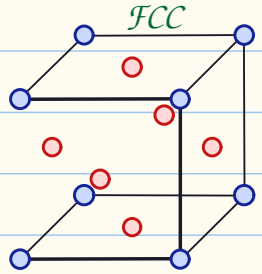
Mixed dislocation & dislocation loops

- A curved dislocation line has edge character where $b \perp \xi$ and screw where $b \parallel \xi$; in between, mixed. The Burgers vector is the same everywhere along the line (a dislocation cannot end inside a crystal – it ends on a surface, boundary, another dislocation (nodes) or closes into a **loop**).
- **Node rule** : at a junction $\Sigma\, b_{in} = \Sigma\, b_{out}$ (like Kirchhoff's current law).
- **Glide loop** : b lies in the plane of the loop – expands/contracts by slip. **Prismatic loop** : $b \perp$ loop plane – formed by condensation of vacancies (vacancy loop) or interstitials (interstitial loop) ; moves only by climb (sessile). **Frank loop** : faulted loop in FCC ($b = a/3\langle 111\rangle$).
- **Dislocation density** ρ : annealed crystal 10^{6}–10^{8} cm^{-2} (10^{10}–10^{12} m^{-2}) ; heavily cold worked 10^{11}–10^{12} cm^{-2} ; ideal (whiskers) 0.

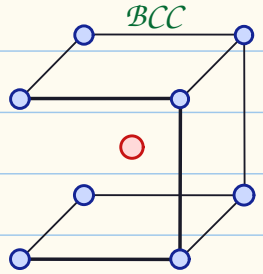
Characteristics summary

Property	Edge	Screw
b vs line ξ	$b \perp \xi$	$b \parallel \xi$
Glide plane	unique ($b \times \xi$)	any plane containing ξ
Direction of motion	$\parallel$ to b	$\perp$ to b (line moves $\perp$ to the shear)
Climb	possible	not possible
Hydrostatic stress field	yes	no (pure shear)
Cross-slip	no	yes

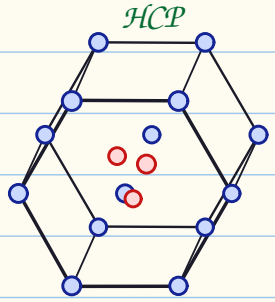
5. Dislocations in FCC, BCC and HCP Lattices



$\{111\}\langle 110\rangle$: 12 systems
$|b| = a/\sqrt{2} = 2.55$ Å (Cu)



$\{110\}\langle 111\rangle$ (+$\{112\},\{123\}$) : 48
$|b| = a\sqrt{3}/2 = 2.48$ Å (Fe)



basal $(0001)\langle 11\text{-}20\rangle$: 3 systems
$|b| = a$ (Zn 2.66 Å)

- **Slip occurs on close-packed planes along close-packed directions** (shortest b ⇒ lowest energy ∝ b^2). Slip system = {plane}⟨direction⟩.

Structure	Slip system	No.	Burgers vector ; length
FCC (Cu, Al, Ni, Au)	{111}⟨110⟩	12	a/2⟨110⟩ ; a/√2
BCC (Fe, W, Cr, Mo)	{110},{112},{123}⟨111⟩	12 + 12 + 24 = 48	a/2⟨111⟩ ; a√3/2
HCP (Zn, Mg, Cd, Ti)	basal (0001)⟨11-20⟩ ; prismatic, pyramidal	3 (basal) + 3 + 6	a/3⟨11-20⟩ ; a (also ⟨c+a⟩)

- **FCC** : 4 {111} planes × 3 ⟨110⟩ directions = 12 systems – very ductile, low CRSS, little temperature dependence. Cu a = 3.615 Å ⇒ $|b|$ = 2.556 Å.
- **BCC** : no truly close-packed plane ; {110} is the densest ; many systems of comparable spacing ⇒ pencil-glide ; CRSS strongly depends on temperature and strain rate (screw dislocations have non-planar cores – high Peierls stress) ⇒ ductile–brittle transition. Fe a = 2.866 Å ⇒ $|b|$ = 2.482 Å.
- **HCP** : only 3 independent basal systems (c/a > 1.63 : Zn, Cd – basal slip, brittle ; c/a ≈ 1.59 Ti, Zr – prismatic slip also ; Mg at room T basal only, non-basal above 225 °C) ⇒ limited ductility, twinning is important.

Partial dislocations & stacking faults (FCC)

- A perfect dislocation in FCC dissociates into two **Shockley partials** joined by a ribbon of stacking fault (…ABCABABC…) :

$a/2[1\,{-1}\,0] \rightarrow a/6[2\,{-1}\,{-1}] + a/6[1\,{-2}\,1]$ (energy lowered since $b_1^2 > b_2^2 + b_3^2$ – Frank's criterion : $a^2/2 > a^2/3$)

- Equilibrium width of the fault (balance of repulsion between partials and fault energy γ) :

$$d \approx G\, b_1 \cdot b_2 / (2\pi\, \gamma_{SF}),\ b_1 \cdot b_2 = a^2/12 \text{ (screw)} ;\ \text{edge : divide by } (1-\nu)$$

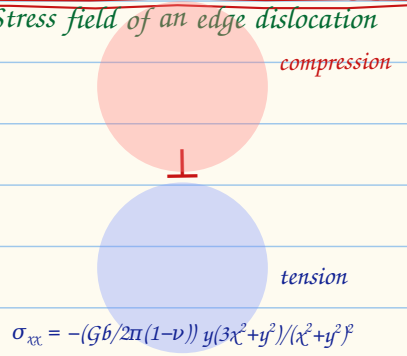
- Stacking-fault energy γ (mJ/m²) : Al 166, Ni 125, Cu 45, Au 32, Ag 22, brass/stainless < 20. High γ (Al) : narrow ribbon, easy cross-slip, easy recovery. Low γ (Cu-alloys, stainless steel) : wide ribbon, difficult cross-slip, rapid work hardening, twinning.

Ex 3. Cu : $a = 3.615$ Å, $G = 48$ GPa, $\gamma = 45$ mJ/m² : $b_1 \cdot b_2 = (3.615\times10^{-10})^2/12 = 1.09\times10^{-20}$ m² ⇒ $d = (48\times10^9 \times 1.09\times10^{-20})/(2\pi \times 0.045) = 1.85$ nm (screw) ; edge $d = 1.85/0.66 = 2.8$ nm (≈ 7–11 atomic spacings).

- **Sessile dislocations** : Lomer–Cottrell lock (formed by reaction of partials on intersecting {111} planes, a/6⟨110⟩ stair-rod) – barrier to glide, strong work hardening in FCC ; Frank sessile loops.
- **Dislocation reactions** : allowed if Σ b conserved and energy (b^2) decreases : e.g. a/2[101] + a/2[01-1]... → a/2[110]. BCC : a/2[111] + a/2[1-1-1] → a[100] (cleavage crack nucleation, Cottrell).

6. Stress Fields & Energy of Dislocations

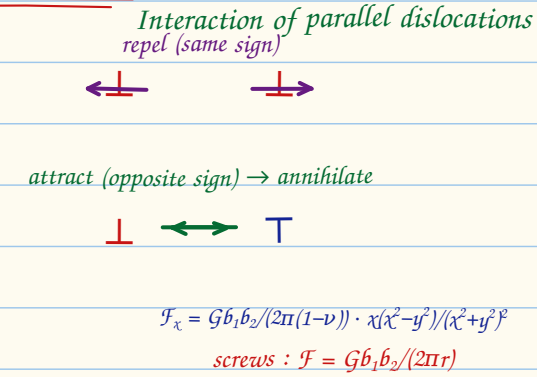
Stress field of an edge dislocation



$\sigma_{xx} = -(Gb/2\pi(1-\nu))\, y(3x^2+y^2)/(x^2+y^2)^2$

$\tau_{xy} = (Gb/2\pi(1-\nu))\, x(x^2-y^2)/(x^2+y^2)^2$

Interaction of parallel dislocations



$F_x = Gb_1b_2/(2\pi(1-\nu)) \cdot x(x^2-y^2)/(x^2+y^2)^2$

screws : $F = Gb_1b_2/(2\pi r)$

Stress field (isotropic elastic continuum)

- **Screw** (cylindrical coordinates, no normal stresses) : $\tau_{\theta z} = G\,b\,/\,(2\pi\,r)$ — varies as 1/r, singular at the core (valid for $r > r_0 \approx b \Rightarrow 2$–$3b$).
- **Edge** (x along b, y normal to the glide plane) :

$$\sigma_{xx} = -\,[Gb/2\pi(1-\nu)]\; y(3x^2+y^2)/(x^2+y^2)^2 \quad \sigma_{yy} = [Gb/2\pi(1-\nu)]\; y(x^2-y^2)/(x^2+y^2)^2$$

$$\tau_{xy} = [Gb/2\pi(1-\nu)]\; x(x^2-y^2)/(x^2+y^2)^2 \quad \sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$$

- Features : compressive σ_{xx} above the glide plane ($y > 0$), tensile below ; τ_{xy} is maximum on the glide plane ($y = 0$) and zero at $x = 0$ and on the lines $x = \pm y$; the stress decays as 1/r – a long-range field, so dislocations interact over large distances.

Strain energy per unit length

- Energy in the elastic field between r_0 (core) and R (outer radius ≈ half the spacing between dislocations ≈ $1/(2\sqrt{\rho})$) :

$$E_{screw} = (G\,b^2/4\pi)\,\ln(R/r_0)\quad E_{edge} = (G\,b^2/4\pi(1-\nu))\,\ln(R/r_0)$$

- Putting $\ln(R/r_0) \approx 4\pi$ ($R \approx 10^3 b$) : $E \approx \alpha\, G\, b^2$, $\alpha = 0.5 - 1$ (J/m) ; **line tension** $T \approx \tfrac{1}{2}\, G\, b^2$ (resists bending – like a stretched rubber string). Edge energy is ≈ 1.5 times that of the screw $(1/(1-\nu))$.
- Energy $\propto b^2 \Rightarrow$ dislocations with the smallest b are most stable ; a dislocation tends to be straight (short) to minimise energy. **Core energy** ≈ $Gb^2/10$ (adds 10–20 %).

Ex 4. Cu : G = 48 GPa, b = 2.55 Å : $Gb^2 = 48\times10^9 \times 6.5\times10^{-20} = 3.12\times10^{-9}$ J/m $\Rightarrow E \approx 0.5\, Gb^2 = 1.56\times10^{-9}$ J/m $\Rightarrow$ per atom length (b) = $1.56\times10^{-9} \times 2.55\times10^{-10} = 4.0\times10^{-19}$ J = 2.5 eV – large compared with kT (0.025 eV) $\Rightarrow$ dislocations are not thermodynamically stable (cannot be removed by cooling) but are **non-equilibrium** defects.

- **Peierls–Nabarro stress** (lattice friction) : $\tau_p = [2G/(1-\nu)]\, \exp[\,-2\pi\, w/b\,]$, $w = a/(1-\nu)$ = width of the dislocation core. Wide core (FCC metals) $\Rightarrow$ small τ_p ; narrow core (covalent, BCC screw) $\Rightarrow$ large τ_p.

7. Forces on and between Dislocations

Force on a dislocation (Peach–Koehler)

An applied shear stress τ on the slip plane in the direction of b exerts a force per unit length of line, perpendicular to the line :

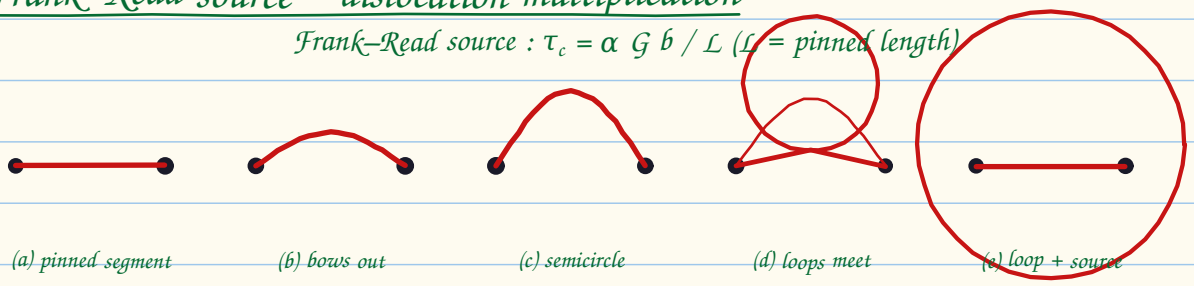
$$F = \tau\, b \text{ (per unit length) ; vector form } \mathbf{F} = (\boldsymbol{\sigma} \cdot \mathbf{b}) \times \boldsymbol{\xi}$$

- Derivation (work) : the slip of the area $\ell \cdot dx$ gives displacement b over area $\ell\, dx$ $\Rightarrow$ work = $\tau(\ell\, dx)\, b = F\, dx \Rightarrow F = \tau\, b\, \ell$. Direction – always normal to the dislocation line, toward the unslipped region.
- **Bowing** : a segment of length L pinned at both ends bows to a radius R under τ : $\tau\, b = T/R \Rightarrow R = Gb/(2\tau)$ (using $T = \tfrac{1}{2}Gb^2$). Minimum radius $L/2 \Rightarrow$ critical stress to operate a source :

$$\tau_c = G\,b / L$$

Frank–Read source – dislocation multiplication

Frank–Read source : $\tau_c = \alpha\, G\, b / L$ (L = pinned length)



- A segment between two pinning points (nodes, precipitates) bows out under stress (a), (b) ; at the semicircle (c) stress is maximum ($R = L/2$, $\tau_c = Gb/L$) ; beyond it the loop expands unstably, parts meet (d) and annihilate to form a complete loop and regenerate the original segment (e) – one source emits many loops ⇒ rapid increase of ρ ($10^6 \rightarrow 10^{12}$ cm^{-2}) during plastic deformation and cold working.

Ex 5. Cu, $L = 1\ \mu m$: $\tau_c = (48\times10^{9})(2.55\times10^{-10})/10^{-6} = 12.2$ MPa ; for $L = 10\ \mu m$: 1.2 MPa. (Shows shorter segments need higher stress – source hardening.)

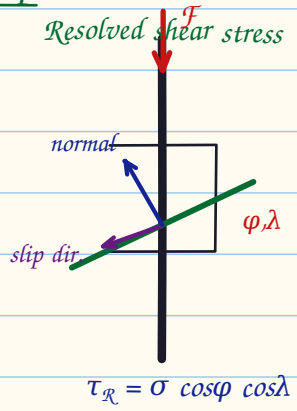
Interaction between parallel dislocations

- **Parallel screws** (b_1, b_2, distance r) : $F = G\, b_1 b_2/(2\pi\, r)$ per unit length – repulsive for like sign, attractive for opposite.
- **Parallel edges on the same glide plane** (separation x) : $F = G\, b_1 b_2/(2\pi(1-\nu)\, x)$ – like signs repel, unlike signs attract and annihilate when they meet (vanish if on the same plane).
- **Parallel edges on different planes** (offset y) : $F_x = [G\, b_1 b_2/2\pi(1-\nu)] \cdot x(x^2 - y^2)/(x^2 + y^2)^2$. Like edges are stable (a stable low-energy configuration, **tilt/small-angle boundary**) at $x = 0$ – one directly above the other ($\perp$ above $\perp$: polygonisation).
- Consequences : (i) **pile-up** at a barrier (grain boundary) – n dislocations with stress concentration $\tau' = n\tau$ ($n = \pi\tau\, L/(G\, b)$ ...) leads to slip transmission (Hall–Petch, Unit II) or crack nucleation ; (ii) interactions with forest dislocations ⇒ work hardening $\tau = \tau_0 + \alpha\, G\, b\, \sqrt{\rho}$ ($\alpha \approx 0.2$–0.5) ; (iii) dislocation–solute (Cottrell atmospheres), dislocation–precipitate interactions ⇒ strengthening.

Ex 6. Cu, $\rho = 10^{15}\ m^{-2}$ (cold-worked) : Taylor $\Delta\tau = 0.5 \times 48\times10^{9} \times 2.55\times10^{-10} \times \sqrt{(10^{15})} = 0.5 \times 12.24 \times 3.16\times10^{7} =$ 193 MPa.

8. Slip & Twinning

Slip



Schmid's law : slip starts when $\tau_R = \tau_{CRSS}$

Schmid factor $m = \cos\varphi \cos\lambda \le 0.5$ ($\varphi=\lambda=45°$)

$\sigma_y = \tau_{CRSS}/m$ (single crystal)

CRSS (MPa) : Cu 0.6–1 ; Al 0.8 ; Zn 0.18 ; Fe ≈ 30 ; Mg 0.8

slip planes : most densely packed ; directions : closest packed

multiple slip → work hardening (stage II)

- **Slip** = shear of one part of the crystal over another along a crystallographic **slip plane** in a **slip direction** by the glide of dislocations. Visible as slip lines/bands (each step ≈ 100–1000 atomic spacings). Slip distance is an integral multiple of the lattice repeat.
- **Schmid's law** : slip begins when the resolved shear stress on the slip system reaches a critical value (CRSS) — a material constant at a given T :

$$\tau_R = (P/A)\cos\varphi\cos\lambda = \sigma\cos\varphi\cos\lambda \quad \sigma_y = \tau_{CRSS}/(\cos\varphi\cos\lambda)$$

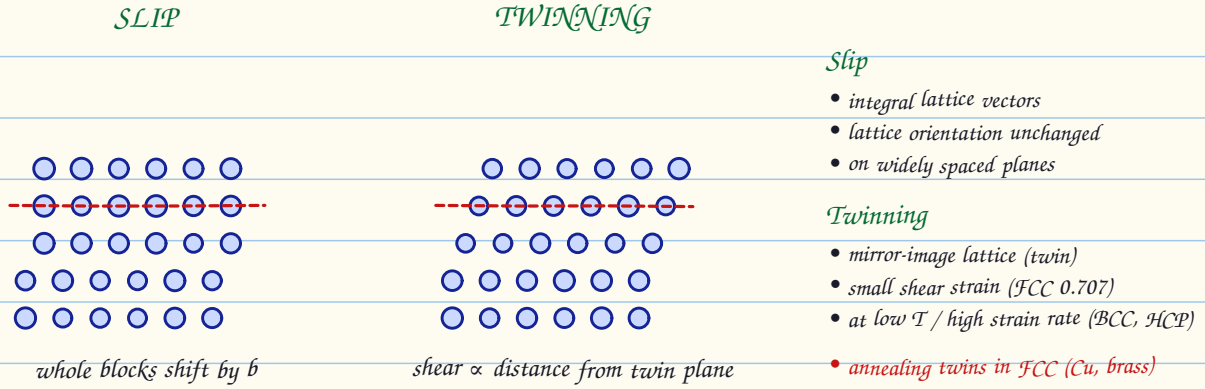
where φ = angle between the tensile axis and the slip-plane normal, λ = angle between the tensile axis and the slip direction ($\varphi + \lambda$ need not be 90°). **Schmid factor** $m = \cos\varphi\cos\lambda \le 0.5$.

- The system with the highest m slips first (primary system). Ex : Zn single crystal yields at very low σ when $m \approx 0.5$, but is hard when the basal plane is parallel ($\varphi = 0°$) or perpendicular ($\lambda = 90°$) to the axis. In polycrystals the average $m \approx 0.31$ (Taylor factor ≈ 3.06 for FCC).

Ex 7. Cu single crystal : $\varphi = 45°$, $\lambda = 35°$, applied $\sigma = 2$ MPa : $m = \cos45° \cos35° = 0.7071 \times 0.8192 = 0.579 \Rightarrow \tau_R = 2 \times 0.579 =$ 1.16 MPa (> CRSS 0.6–1 MPa ⇒ slips). The tensile yield stress for this orientation = 0.98/0.579 = 1.7 MPa.

- Effects of slip : crystal rotation toward the tensile axis (slip direction aligns with load), lattice curvature, work hardening (stage I easy glide, II linear hardening, III dynamic recovery by cross-slip). **Slip bands** may be coarse (wavy vs planar slip depending on γ_{SF}).
- **Kink bands** : localised bending (hcp crystals loaded parallel to basal plane).

Twinning



- **Twinning** = a homogeneous shear of a region of the crystal such that the deformed region is the mirror image of the parent across the **twin plane** ; every atomic plane moves by an amount ∝ its distance from the twin plane (fraction of a lattice vector) – unlike slip.
- Twin systems : FCC {111}⟨11-2⟩ (shear $s = 1/\sqrt{2} = 0.707$; by passage of a/6⟨112⟩ Shockley partials on successive planes) ; BCC {112}⟨111⟩ ($s = 0.707$) ; HCP {10-12}⟨10-11⟩ (Zn, Mg, Ti).

Feature	Slip	Twinning
Lattice orientation	unchanged (steps)	reoriented (mirror image)
Shift	integral lattice vectors	fraction of lattice vector
Occurs on	widely spaced planes	every plane in the twinned region
Strain produced	large	small (< 10 %)
Conditions	any temperature	low T, high strain rate, few slip systems (BCC, HCP) ; FCC low γ
Rate	moderate	very fast, audible "tin cry"
Tool	dislocation glide	twinning dislocation / pole mechanism

- Types : **mechanical (deformation) twins** (BCC Neumann bands in Fe, HCP) and **annealing twins** (FCC Cu, brass – formed during recrystallisation, straight parallel-sided). Roles : reorient crystals so that further slip becomes possible ; accommodate strain in HCP ; trigger cleavage cracks in BCC at low T. Twinning stress is relatively temperature independent (stress to nucleate high, to propagate low).

9. Additional Solved Problems

P1. A tensile test on a steel bar of 12 mm dia and gauge length 50 mm : load at yield 56.5 kN, fracture dia 7 mm, maximum load 75 kN. $A_0 = 113.1\ mm^2 \Rightarrow \sigma_y = 56500/113.1 = 500\ MPa$; $UTS = 75000/113.1 = 663\ MPa$; $\%\ RA = (113.1 - 38.5)/113.1 \times 100 = 66\ \%$.

P2. True strain from engineering strain 0.20 : $\varepsilon_T = \ln 1.2 = 0.182$; $\sigma = 300\ MPa \Rightarrow \sigma_T = 360\ MPa$.

P3. Burgers vector of Al ($a = 4.05$ Å) : $|b| = a/\sqrt{2} = 2.86$ Å ; Mo ($a = 3.15$ Å, BCC) : $|b| = 3.15 \times 0.866 = 2.73$ Å.

P4. Force per unit length on a dislocation (Cu, $b = 2.55$ Å) under $\tau = 10\ MPa$: $F = \tau\, b = 10\times10^{6} \times 2.55\times10^{-10} = 2.55\times10^{-3}\ N/m$.

P5. Dislocation density $10^{10}\ m^{-2} \Rightarrow$ spacing $\approx 1/\sqrt{\rho} = 10\ \mu m$; at $10^{15}\ m^{-2} \Rightarrow 0.03\ \mu m$ (30 nm).

Short answers – Quick Revision

- *Why do metals yield far below the theoretical shear strength?* Plastic flow occurs by glide of dislocations, breaking only a few bonds at a time.
- *Is E changed by cold working?* No (structure-insensitive) ; yield and tensile strength are structure-sensitive.
- *Why can screw dislocations cross-slip but edge cannot?* $b \parallel \xi$ means no unique glide plane for the screw.
- *Why is FCC more ductile than HCP?* 12 equivalent slip systems versus only 3 basal.
- *Why are dislocation energies proportional to b^2?* Elastic strain energy $\propto$ (strain)^2, strain $\propto b$.

10. Quick Formula Sheet – Unit I

Item	Formula
Hooke's law	$\sigma = E\varepsilon$; $\tau = G\gamma$; $E = 2G(1+\nu) = 3K(1-2\nu)$
True stress/strain	$\sigma_T = \sigma(1+\varepsilon)$; $\varepsilon_T = \ln(1+\varepsilon)$
Theoretical strength	$\tau_{th} \approx G/2\pi$
Screw stress	$\tau_{\theta z} = Gb/2\pi r$
Energy per length	$E \approx \frac{1}{2}Gb^2$; screw $Gb^2/4\pi \cdot \ln(R/r_0)$
Force on dislocation	$F = \tau b$
Frank–Read	$\tau_c = Gb/L$
Taylor hardening	$\Delta\tau = \alpha G b \sqrt{\rho}$
Schmid's law	$\tau_R = \sigma \cos\varphi \cos\lambda$
Partial separation	$d = Gb_1 \cdot b_2/(2\pi\gamma)$

11. Important Questions

- State Hooke's law; derive the relation between E, G, K and ν. Explain the origin of elastic behaviour from the interatomic force curve.
- Why is the observed yield strength much lower than the theoretical? Explain the need for dislocations.
- Differentiate edge and screw dislocations ; define Burgers vector and Burgers circuit ; explain dislocation loops.
- Describe dislocations in FCC, BCC and HCP lattices; Burgers vectors; partial dislocations & stacking fault.
- Derive the stress field and energy of a screw dislocation; explain forces between dislocations; Frank–Read source.
- State Schmid's law with derivation. Compare slip and twinning.

UNIT – II

Strengthening Mechanisms

Subject : Mechanical Behaviour of Materials (23A03504c)

> Syllabus: Cold working, grain-size, solid-solution, martensitic, precipitation, dispersion and fibre strengthening with examples ; yield-point phenomenon, strain ageing and dynamic strain ageing.

1. Principle of Strengthening

Plastic deformation occurs by dislocation motion – so strengthening = restricting dislocation motion. Every mechanism introduces obstacles that raise the stress needed to move dislocations ($\tau = \tau_0 + \Delta\tau$), at the expense of ductility and toughness (usually).

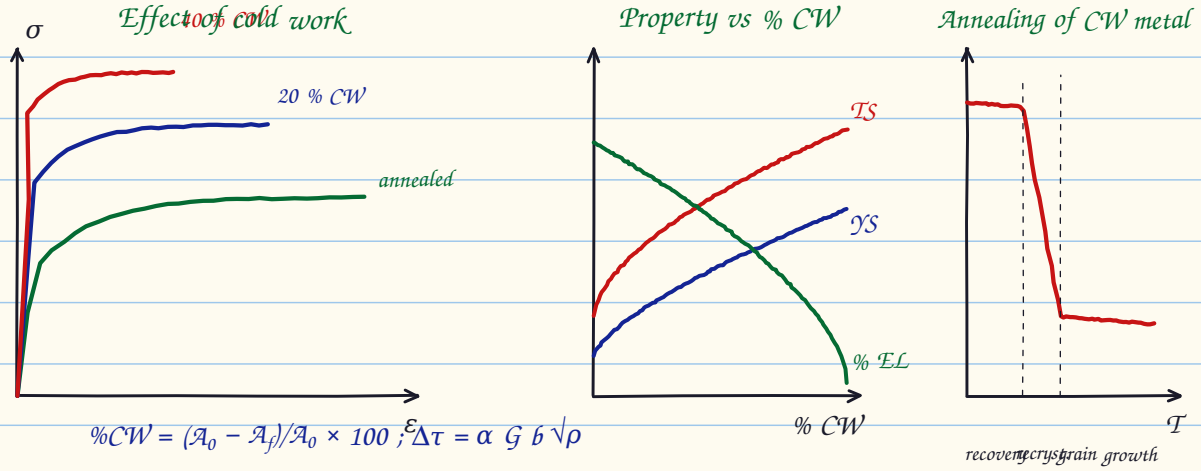
Obstacle	Mechanism	Technique
Other dislocations	forest hardening	cold work
Grain boundaries	pile-up blocked	grain refinement
Solute atoms	lattice friction / atmospheres	alloying
Fine precipitates	shear or bypass	age hardening
Hard dispersoids	Orowan bypass	powder metallurgy (ODS)
Fibres / reinforcement	load transfer	composites
Phase / BCT lattice	supersaturation, high ρ	quench (martensite)

- Strengthening effects are roughly additive : $\sigma_y = \sigma_0 + \Delta\sigma_{ss} + \Delta\sigma_{gb} + \Delta\sigma_\rho + \Delta\sigma_{ppt}$ (sometimes by root-sum-square).
- Requirement : a good strengthening process should leave ductility, toughness and high-temperature stability acceptable.

2. Cold Working (Strain Hardening / Work Hardening)

Plastic deformation below the recrystallisation temperature ($\approx 0.4\ T_m$) – rolling, drawing, forging, bending.

$$\%\ CW = (A_0 - A_f)/A_0 \times 100$$



- ***Mechanism*** *: plastic strain multiplies dislocations (Frank–Read sources) – ρ rises from 10^6–10^8 to 10^{11}–10^{12} cm^{-2} ; dislocations interact, tangle, form pile-ups, cells and locks (Lomer–Cottrell), so further slip needs a higher stress.*

$$\Delta\tau = \alpha\ G\ b\ \sqrt{\rho}\ \text{(Taylor, } \alpha \approx 0.2\text{–}0.5) \quad \sigma_T = K\varepsilon_T^{\,n}$$

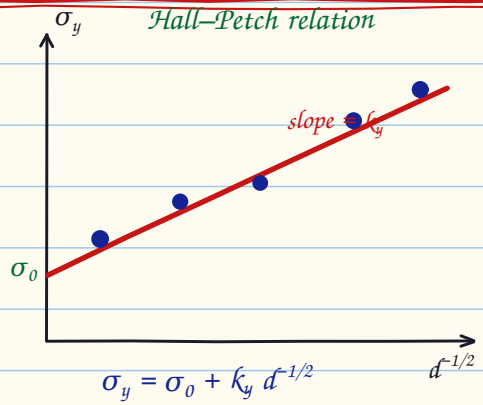
- *Effects of cold work : YS ↑↑, TS ↑, hardness ↑, ductility ↓, electrical conductivity slightly ↓, corrosion (stress) susceptibility ↑,* ***anisotropy*** *(texture, banding), stored energy ≈ 5 – 10 % of the work retained.*
- *Strain hardening exponent n : FCC (Cu, brass, stainless) n = 0.3–0.5 ; low-C steel 0.2 ; HCP 0.1–0.2. Stage hardening of single crystals : I easy glide, II linear (θ ≈ G/300), III parabolic (dynamic recovery).*
- ***Limit*** *: hardening cannot be increased indefinitely (ductility exhausted) ; it is lost on annealing (recovery → recrystallisation → grain growth).* ***Recrystallisation temperature*** *≈ 0.3–0.5 T_m ; decreases with % CW, purity, small prior grain size. Typical : pure Cu 200 °C, Fe 450 °C, Al 80 °C, W 1200 °C.*

Ex 1. Wire drawn from 5 mm to 4 mm dia : A_0 = 19.63, A_f = 12.57 mm^2 ⇒ % CW = 36 %. For ρ rising to 10^{15} m^{-2} in Cu, $\Delta\tau = 0.5 \times 48\times10^{9} \times 2.55\times10^{-10} \times 3.16\times10^{7}$ = 193 MPa.

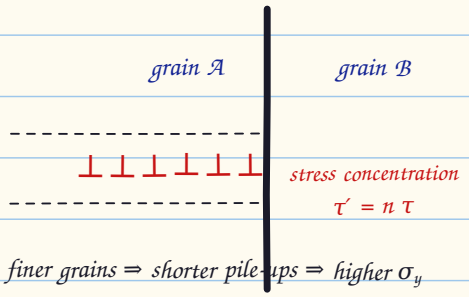
- *Applications : cold-drawn wires, cold-rolled sheet (strain-hardened tempers H12–H18 Al), copper conductors, cold-worked stainless steel springs, shot peening (surface).*

3. Grain-Size Strengthening (Hall–Petch)

Hall–Petch relation



Pile-up at a grain boundary



Grain boundaries stop slip (adjacent grain has a different orientation) – dislocations pile up against them. The stress concentration of n piled-up dislocations ($\tau' = n\tau$) must reach the stress needed to activate a source in the next grain :

$$\sigma_y = \sigma_0 + k_y\, d^{-1/2} \quad \text{(Hall 1951, Petch 1953)}$$

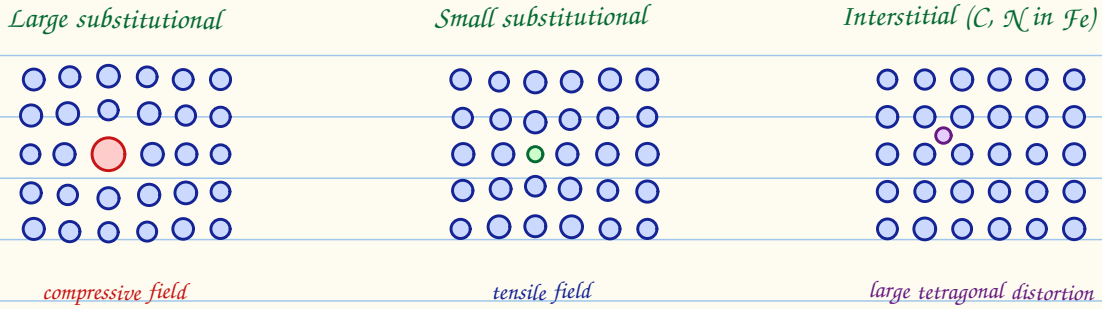
- σ_0 = friction stress (lattice resistance to dislocation motion) ; k_y = locking parameter (measure of how effectively grain boundaries block slip) ; d = average grain diameter. Valid for d from ≈ 1 mm to ≈ 20 nm.
- k_y ($MPa \cdot m^{1/2}$) : low-C steel ≈ 0.6 ; Cu 0.11 ; Al 0.07 ; Ti 0.4 ; Mg 0.28 ; Ni 0.16. Increase of k_y for BCC (strong locking by atmospheres).
- Fine grain also improves toughness and lowers DBTT (unlike other methods) – so the best way to raise strength and toughness together (HSLA steels, controlled rolling). Grain refinement methods : rapid solidification, inoculation (Ti-B in Al), recrystallisation after cold work, thermo-mechanical controlled processing, microalloying (Nb, V, Ti), severe plastic deformation (ECAP).
- ***Inverse Hall–Petch*** : for d < 10–20 nm (nano-crystals) strength falls – grain-boundary sliding dominates. ***Limits*** : at high temperature (> 0.5 T_m) fine grains are weaker (grain-boundary sliding, creep – see Unit V).

Ex 2. Low-carbon steel σ_0 = 100 MPa, k_y = 0.6 $MPa \cdot m^{1/2}$: d = 25 µm ⇒ $d^{-1/2}$ = 200 $m^{-1/2}$ ⇒ σ_y = 100 + 0.6×200 = 220 MPa ; d = 4 µm ⇒ 500 ⇒ 400 MPa ; d = 1 µm ⇒ 700 MPa.

- ASTM grain size : $N = 2^{n-1}$ grains per square inch at 100× ; larger n ⇒ finer grains.

4. Solid-Solution Strengthening

Solute atoms in the matrix (substitutional or interstitial) make the alloy stronger than the pure metal because they distort the lattice and interact with dislocations.



- **Sources of interaction** : (1) size misfit $\varepsilon = (1/a)(da/dc)$ – strain field interacts with the hydrostatic part of a dislocation's field (edge) ; (2) modulus mismatch ; (3) electrical / chemical (Suzuki segregation to stacking faults) ; (4) order effects. Large solutes sit in the tensile region below an edge dislocation, small ones in the compressive region above.
- **Cottrell atmosphere** : solute accumulates around dislocation cores (C, N in Fe ; interstitials) and lowers their energy by ≈ 0.5 eV — the dislocation is locked ⇒ higher stress to break away (upper yield point, strain ageing).

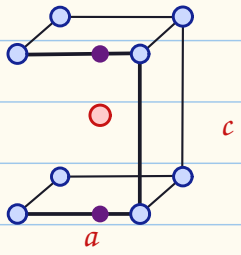
$\Delta\tau \propto G\,\varepsilon^{3/2}\,c^{1/2}$ (Labusch/Fleischer, dilute) ; or $\Delta\tau \propto c^{2/3}$; $\Delta\tau = G\,\varepsilon\,c$ (Mott–Nabarro)

- Strengthening increases with : solute concentration c ; misfit ε (Hume-Rothery rule : size difference < 15 % for extensive solubility) ; interstitials are 10–100 times as effective as substitutional (non-spherical distortion, tetragonal field interacts with shear stress of screw also).
- Examples : brass (Cu–Zn), bronze (Cu–Sn), Cu–Ni (monel), Al–Mg (5xxx alloys), stainless steel (Cr, Ni, N), Mn/Si in ferrite. Approx. rise of ferrite yield strength per wt % : Si ≈ 80 MPa ; Mn ≈ 30 MPa ; Cu ≈ 40 MPa ; P ≈ 700 MPa ; interstitial C+N ≈ 4000 MPa/wt % (limited by solubility).
- Effect on properties : YS, TS ↑, ductility ↓, conductivity ↓ strongly, annealed alloys work-harden faster. Maximum strengthening near 50 % composition for complete-solubility alloys (Cu–Ni). Temperature : solute strengthening persists to high T (≈ 0.6 T_m) – basis of solid-solution superalloys (Inconel 625).

Ex 3. Cu has YS ≈ 70 MPa ; 70-30 brass (30 % Zn) annealed has YS ≈ 100 MPa, TS 325 MPa ; Cu + 10 % Ni annealed TS ≈ 270 MPa and Cu + 30 % Ni ≈ 380 MPa.

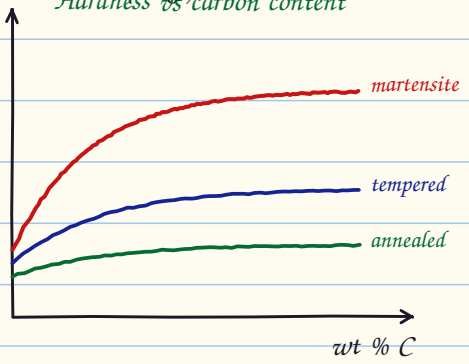
5. Martensitic Strengthening

Body-centred tetragonal (martensite)



$c/a = 1 + 0.045$ (wt % C) ; C at octahedral sites

Hardness vs carbon content



Martensite forms when austenite (FCC γ-Fe containing carbon) is cooled so rapidly (quenched) that carbon cannot diffuse out as cementite ; the lattice shears (diffusionless, athermal, ≈ speed of sound) into a body-centred tetragonal (BCT) structure supersaturated with carbon.

- **Sources of strength** : (1) interstitial carbon – very potent solid-solution hardening ; (2) very high dislocation density (10^{11}–10^{12} cm^{-2}) and twins formed by the shear transformation ; (3) extremely fine lath/plate substructure (boundaries – Hall–Petch) ; (4) carbon segregation to dislocations and later fine carbide precipitates (in tempering).

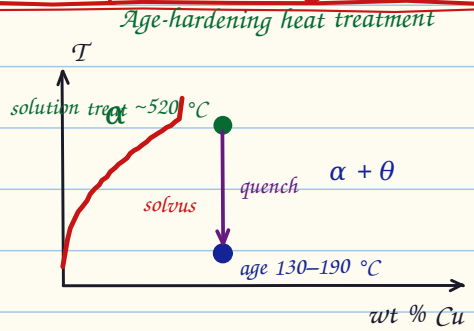
$$c/a = 1 + 0.045 \text{ (wt \% C)} \quad M_s(°C) = 539 - 423\,C - 30.4\,Mn - 17.7\,Ni - 12.1\,Cr - 7.5\,Mo$$

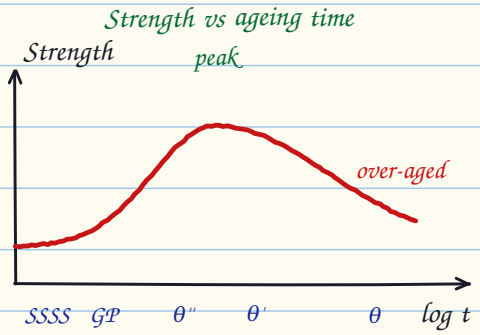
- Morphology : lath (low C < 0.6 %, high dislocation density, tougher) ; plate (high C > 1 %, twinned, brittle, many micro-cracks) ; mixed between. Retained austenite for M_f < room T (high C).
- Hardness of martensite depends mainly on carbon : HRC ≈ 35 (0.2 % C), 55 (0.4 %), 65 (≥ 0.6 %). Yield strength may be 1500–2000 MPa but brittle ⇒ tempering (150–650 °C) – carbon precipitates as carbides (ε-carbide → Fe_3C), lath recovers : strength ↓, toughness ↑ (tempered martensite = best combination).
- Hardenability (depth) governed by alloy content (Mn, Cr, Mo, Ni) – Jominy test ; quenching cracks and distortion are limitations. Related : bainite (carbide +

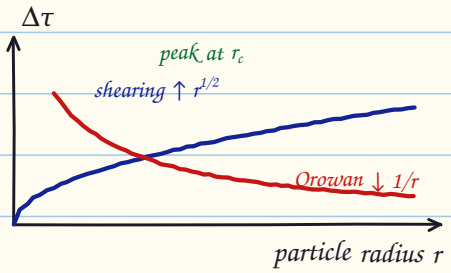
ferrite, ≈ 0.5 hardness), **TRIP** and dual-phase steels, maraging steels (Fe–18Ni martensite aged with Ni_3Ti, 2000 MPa).

Ex 4. 0.4 % C, 0.8 % Mn steel : $M_s = 539 - 169.2 - 24.3 = 345\ °C$; $c/a = 1 + 0.045 \times 0.4 = 1.018$.

6. Precipitation (Age) Hardening







SSSS → GP zones → θ″ → θ′ → θ (Al_2Cu)

coherent → semi-coherent → incoherent

Alloys whose solubility of the solute decreases with temperature can be strengthened by a controlled precipitate dispersion (Wilm 1906 – duralumin). Examples : Al–Cu (2xxx), Al–Mg–Si (6xxx), Al–Zn–Mg (7xxx), Cu–Be, Ni-base superalloys, maraging steel, PH stainless steels (17-4).

Heat treatment steps

- (1) **Solution treatment** : heat into single-phase α region (≈ 520 °C for Al–4.5Cu) and soak to dissolve the second phase. (2) **Quench** rapidly – supersaturated solid solution (SSSS) is retained. (3) **Ageing** : natural (room T) or artificial (130–190 °C) – decomposition proceeds.
- **Precipitation sequence** (Al–Cu) : SSSS → GP zones (coherent Cu-rich discs, 1–2 atom layers) → θ″ (coherent, tetragonal) → θ′ (semi-coherent) → θ (Al_2Cu) (incoherent equilibrium). Strength maximum at θ″ + θ′ ; **over-ageing** (coarse θ) lowers strength.

Strengthening mechanisms

- **Coherency strain hardening** (small zones) : lattice misfit strains impede dislocations.

- **Particle shearing** (small, coherent, $r < r_c$) : dislocations cut through – extra energy for new interface area (surface hardening), order hardening (γ' in Ni alloys), modulus misfit, stacking-fault differences. $\Delta\tau \propto \sqrt{(f\,r)}$ – increases with r.
- **Orowan bypassing** (large, incoherent, $r > r_c$) : dislocation bows between particles and leaves loops :

$$\Delta\tau = G\,b / \lambda \quad \lambda \approx 2r(\sqrt{(\pi/4f)} - 1) \text{ (decreases as particles coarsen)}$$

- Peak strength at the critical radius r_c where the shearing and Orowan stresses are equal.

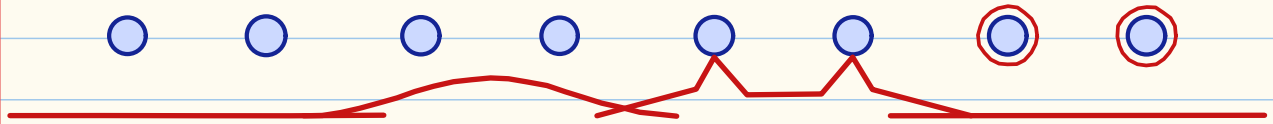
Ex 5. Al alloy : $G = 26$ GPa, $b = 2.86$ Å ($Gb = 7.4$ N/m), $f = 0.02$, $r = 10$ nm : $\lambda = 2(10)(\sqrt{(\pi/0.08)} - 1) = 20 \times 5.27 = 105$ nm $\Rightarrow \Delta\tau = 7.4/105\times10^{-9} =$ 70 MPa $\Rightarrow \Delta\sigma \approx M\,\Delta\tau = 3.06 \times 70 \approx$ 215 MPa. If particles coarsen so $\lambda = 1$ µm : $\Delta\tau =$ 7.4 MPa (over-aged).

- Requirements for an age-hardenable system : appreciable maximum solid solubility, solubility ↓ with T, an alloy that forms coherent transition precipitates. Natural ageing (T4) vs artificial (T6) ; duralumin hardens at RT in days. Dis-advantages : loses strength above ≈ 150–200 °C (coarsening, dissolution) ; welding zones soften (HAZ).
- **Related** : Spinodal decomposition (Cu–Ni–Fe, alnico) ; strain-age (double) treatments ; precipitation hardened Ni-superalloys retain strength to 0.8 T_m via ordered γ' $Ni_3(Al,Ti)$.

7. Dispersion Strengthening

Orowan bowing around strong particles : $\tau = G\,b / \lambda$

loops accumulate ⇒ rapid work hardening



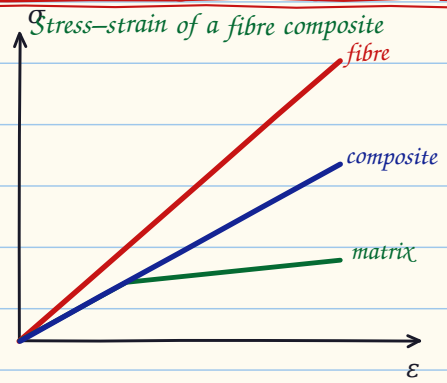
(a) approach (b) bows between (c) pinches off (d) leaves loops

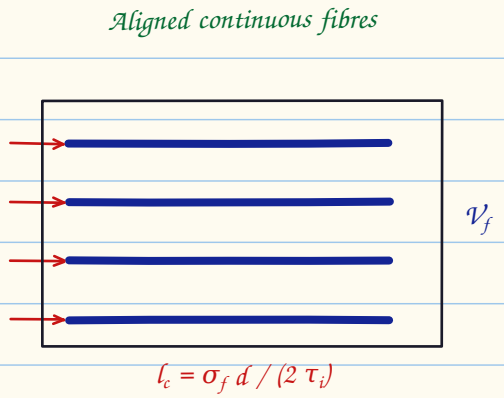
Very fine, stable, insoluble second-phase particles (oxides, carbides, nitrides – Al_2O_3, ThO_2, Y_2O_3) introduced by powder metallurgy / mechanical alloying / internal oxidation / in-situ reaction. Particle size 10–100 nm, volume fraction 1–15 %,

spacing $\lambda \approx 0.1$–$0.5\ \mu m$.

- Strengthening by **Orowan bowing** : $\tau = G\, b / \lambda$; work hardening rate is high (loops left behind).
- **Key difference from age hardening** : particles do not dissolve or coarsen at high temperature (no solubility) ⇒ strength retained up to $\approx 0.9\ T_m$, excellent creep resistance. Strength at room temperature is lower than age-hardened alloys, and ductility/toughness lower.
- Materials : SAP (sintered Al powder with Al_2O_3), TD-nickel (Ni–2 % ThO_2), ODS alloys (MA754, MA6000, PM2000 Fe-Cr-Al–Y_2O_3), dispersion-strengthened copper (Cu–Al_2O_3 "Glidcop" – contact tips, resistance welding electrodes), W–ThO_2 filaments, cemented carbides (WC-Co).
- Conditions for effectiveness : high G of particles, uniform distribution, small size at fixed f ($\lambda \propto r/\sqrt{f}$), particle–matrix interface strong and no reaction ; ratio of Orowan stress to Peierls stress.
- Example : Cu with 0.5 vol % Al_2O_3 ($\lambda \approx 0.1\ \mu m$) has 380 MPa YS with conductivity 90 % IACS and retains it at 900 °C, while age-hardened Cu–Be softens above 300 °C.

8. Fibre Strengthening (Composites)





Strong, stiff fibres embedded in a ductile matrix : the matrix transfers load to the fibres through the interface shear stress, fibres carry most of the load. Fibres : glass, carbon, boron, Kevlar (aramid), SiC, Al_2O_3, steel wire ; matrices : polymers (epoxy), metals (Al, Ti), ceramics.

- **Continuous aligned fibres, load parallel** (isostrain, rule of mixtures) :

$E_c = E_f V_f + E_m V_m$ $\sigma_c = \sigma_f V_f + \sigma_m' V_m$ (σ_m' = matrix stress at fibre-fracture strain)

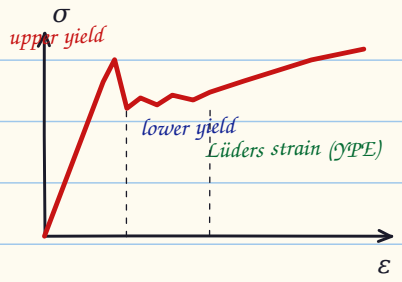
- **Transverse load** (isostress) : $1/E_c = V_f/E_f + V_m/E_m$ – much lower ; strongly anisotropic ⇒ use cross-ply laminates.
- **Critical fibre length** (fibre fully loaded to its strength) : $l_c = \sigma_f d / (2 \tau_i)$. Fibres with $l < l_c$ do not reach fracture stress ; discontinuous (short) fibres : $\sigma_c = \sigma_f V_f(1 - l_c/2l) + \sigma_m' V_m$ for $l > l_c$; random orientation gives ≈ 0.2–0.4 of the aligned value.
- **Matrix role** : bonding, load transfer, protects fibres, separates them (crack stopper) ; ductile matrix with brittle fibres gives toughness through fibre pull-out and debonding.

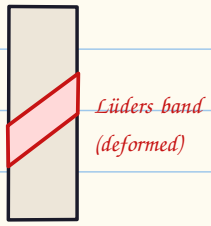
Ex 6. Glass/epoxy, $V_f = 0.5$: $E_f = 72$, $E_m = 3$ GPa ⇒ E_c (parallel) = 36 + 1.5 = 37.5 GPa ; transverse : $1/E = 0.5/72 + 0.5/3 = 0.174$ ⇒ 5.8 GPa. $\sigma_f = 2000$ MPa, $\sigma_m' = 60$ MPa : $\sigma_c = 1000 + 30 =$ 1030 MPa. $l_c = (2000 \times 10\ \mu m)/(2 \times 20\ MPa) =$ 0.5 mm.

- Other composite types : particulate (cermets, concrete), laminar (plywood, clad metals), MMC (SiC/Al – engine cylinders), CMC (SiC/SiC – turbine hot parts), carbon/carbon (brakes). Specific strength (σ/ρ) and specific stiffness (E/ρ) are very high – aerospace, sports goods, wind blades.
- **Similar principle in steel** : ferrite-pearlite, cementite lamellae strengthening ; reinforced concrete ; whiskers (SiC).

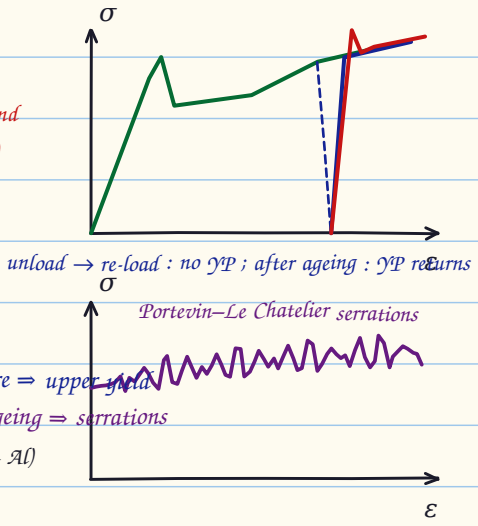
9. Yield Point Phenomenon

Yield-point phenomenon





Strain ageing



Low-carbon steels (and Mo, Ti, Mo alloys with interstitials) show a sharp upper yield point followed by a drop to a lower yield point and plastic flow at nearly constant stress (**yield point elongation**, 1–3 % – Lüders strain) before strain hardening begins.

- **Lüders bands** : localised bands of plastic deformation (≈ 45° to axis) nucleate at stress concentrations and propagate along the gauge length at constant load ; the whole specimen yields when the band front has traversed it. Surface appearance : stretcher-strain marks / "worms" in deep-drawn sheet – unacceptable in automobile panels.
- **Explanation (Cottrell–Bilby)** : C and N atoms segregate to dislocation cores forming atmospheres that anchor them ; a high stress is needed to tear the dislocations away (upper yield) ; once free they move at a lower stress (lower yield). **Johnston–Gilman** : initial mobile dislocation density is very low, so to achieve the imposed strain rate each must move fast ; $v \propto \tau^m$ ($m \approx 20$–30) ⇒ the stress needed is high at first and falls as dislocations multiply ($\dot{\varepsilon} = \rho_m\, b\, v$).
- **Removal** : (i) temper (skin-pass) rolling 0.5–2 % – generates mobile dislocations ; (ii) Al-killed or Ti-stabilised steel (N tied up), IF steel (interstitial-free) ; (iii) forming soon after temper rolling ; (iv) levelling by roller.
- Factors : more pronounced for fine grain, low temperature, high strain rate ; absent in pure FCC metals. Observed in Fe, Mo, Ti, Cd, Zn, Al–Mg, and in single crystals of Ge, Si, LiF.

10. Strain Ageing & Dynamic Strain Ageing

Static strain ageing

- Sequence : (1) strain steel beyond the yield point to A ; (2) unload and **immediately reload** – smooth elastic–plastic curve, no yield point (dislocations free) ; (3) **age** – hold at room temperature for days or at 100–200 °C for minutes – then reload : yield point returns at a higher stress (σ_y ↑, ductility ↓).
- Cause : interstitial C and N diffuse back to the dislocations (pinning, Cottrell) ; Harper's law : fraction pinned $f = 1 - \exp[-(Kt)^{2/3}] \propto (D\,t)^{2/3}$; strain ageing activation energy ≈ that of C diffusion in ferrite (80 kJ/mol).
- **Consequences** : stretcher strains reappear in stored sheet (use within weeks), hardness ↑ and toughness ↓ (↑ DBTT) of cold-worked structural steel after ageing ; **blue brittleness** at 200–300 °C (blue oxide colour) – ductility minimum.

Dynamic strain ageing (DSA)

- Strain ageing that occurs during deformation, when solute diffusion rate is comparable to the speed at which dislocations are arrested at obstacles (waiting time t_w ≈ diffusion time) : occurs in a window of temperature and strain rate (e.g. 100–300 °C for C/N in steel, room temperature for Mg in Al, 400–600 °C for many alloys).
- **Portevin–Le Chatelier (PLC) effect** : repeated yield drops – **serrated stress–strain curve** (types A – periodic locking, B – oscillating, C – sudden drops) due to repeated pinning and unpinning of dislocations ; visible bands in Al–Mg alloys (5083, 5182 – surface roughness "Lüders-like" defects).
- Characteristic signs : negative strain-rate sensitivity ($\partial\sigma/\partial \ln \dot{\varepsilon} < 0$), increased work-hardening rate and flow stress, ductility minimum, strength peak with temperature (bump in σ–T curve), and serrations above a critical strain ε_c that rises with strain rate and falls with temperature.
- Avoid : by changing composition (less Mg/N), heat treatment (precipitation of solutes), avoiding DSA temperature window in forming, texture. Practical : structural steel at ≈ 250 °C (reactor pressure vessel) loses ductility.

11. Examples & Comparison of Mechanisms

Mechanism	Example	Principal obstacle	Limit
Cold work	Cu wire, Al H14 sheet	forest dislocations	annealed away at 0.4 T_m
Grain size	HSLA steel, ultrafine Al	grain boundaries	loses at high T (creep)
Solid solution	brass, stainless, Al–Mg	solute atoms	moderate gain ; < 0.6 T_m
Martensite	quench & tempered 4340	carbon, ρ, laths	brittle ; tempers away ≈ 200 °C
Precipitation	2024-T6, 7075, Cu–Be, Inconel 718	coherent/shearable particles	coarsen/dissolve at ≈ 0.5 T_m
Dispersion	TD-Ni, ODS, SAP, Glidcop	inert oxides (Orowan)	lower RT strength
Fibre	GFRP, CFRP, SiC/Al	fibre load-carrying	anisotropy, cost

- **Combination in practice** : HSLA steel ($\sigma_y \approx 450$ MPa) : $\sigma_0 \approx 50$ + solid solution (Mn, Si) 90 + grain size ($d = 5\ \mu m$) 160 + dislocations 60 + Nb/V carbide precipitates 90 ⇒ ≈ 450 MPa. Ni superalloy : solid solution + γ′ (ordered) + carbides on boundaries + (single crystal, no boundaries).

12. Additional Solved Problems

P1. A Cu rod 12.7 mm dia is cold drawn to 10.2 mm : $\%CW = (126.7 - 81.7)/126.7 = 35.5\ \%$.

P2. Brass : $\sigma_y = 70$ MPa at $d = 0.016$ mm and 140 MPa at $d = 0.0008$ mm ⇒ $d^{-1/2} = 7.9$ and 35.4 $mm^{-1/2}$ ⇒ $k_y = 70/27.5 = 2.55\ MPa{\cdot}mm^{1/2}$; $\sigma_0 = 70 - 2.55 \times 7.9 = 49.9$ MPa.

P3. Martensite M_s for 0.8 % C steel : $539 - 423(0.8) = 200.6$ °C ($M_f \approx -50$ °C ⇒ retained austenite at RT).

P4. Orowan stress for Ni alloy $\lambda = 50$ nm, $G = 76$ GPa, $b = 2.49$ Å : $\Delta\tau = 76\times10^{9}\times2.49\times10^{-10}/(50\times10^{-9}) = 378$ MPa.

P5. Fibre composite : Al–SiC, $V_f = 0.3$, $E_f = 400$, $E_m = 70$ GPa : $E_c = 120 + 49 = 169$ GPa.

Short answers – Quick Revision

- **Which strengthening method does not reduce toughness?** Grain refinement.
- **Why does age hardening peak and then fall?** Shearing → Orowan transition ; over-ageing coarsens particles ($\lambda \uparrow$).
- **Why is carbon more effective than substitutional atoms?** Asymmetric tetragonal distortion ; interacts with both edge and screw dislocations.
- **What causes Lüders bands?** Locally unpinned dislocations multiply in a band ; the band front advances at the lower yield stress.
- **What does a serrated flow curve indicate?** Dynamic strain ageing (PLC effect).

13. Quick Formula Sheet – Unit II

Item	Formula
Cold work	$\%CW = (A_0 - A_f)/A_0 \times 100$; $\Delta\tau = \alpha G b \sqrt{\rho}$
Hall–Petch	$\sigma_y = \sigma_0 + k_y\, d^{-1/2}$
Solid solution	$\Delta\tau \propto c^{1/2}$ (or $c^{2/3}$)
Martensite	$c/a = 1 + 0.045\ \%C$
Orowan	$\Delta\tau = Gb/\lambda$; $\lambda \approx 2r(\sqrt{(\pi/4f)} - 1)$
Composite	$E_c = E_f V_f + E_m V_m$; $l_c = \sigma_f d/2\tau_i$
Strain ageing	$f \propto t^{2/3}$ (Harper)

14. Important Questions

- Explain strain (work) hardening and the effect of cold work on properties ; discuss recovery, recrystallisation and grain growth.
- Derive/Explain the Hall–Petch relationship. Why does grain refinement improve toughness?
- Explain solid-solution strengthening and Cottrell atmosphere. Compare interstitial and substitutional solutes.
- Explain the strengthening of steel by martensite. How does tempering modify it?
- Describe precipitation hardening of Al–Cu with the heat treatment, precipitation sequence and mechanisms (shearing vs Orowan).
- Differentiate dispersion strengthening and age hardening; explain fibre strengthening with the rule of mixtures.

- Explain the yield-point phenomenon, strain ageing and dynamic strain ageing (PLC effect).

UNIT – III

Fracture & Fracture Mechanics

Subject : Mechanical Behaviour of Materials (23A03504c)

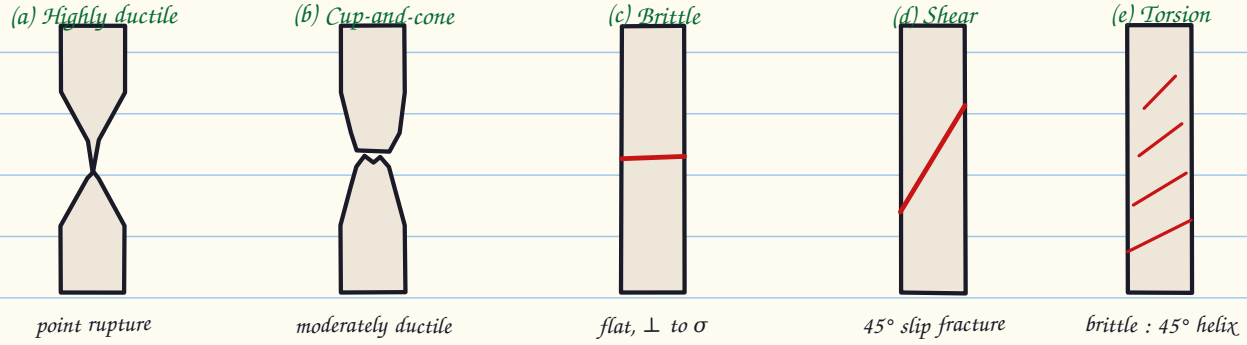
Syllabus: Types of fracture, mechanisms of ductile and brittle fracture, Griffith's theory, ductile-to-brittle transition temperature (DBTT) – factors & determination ; fracture mechanics – modes of fracture, stress intensity factor, strain energy release rate, fracture toughness and determination of K_{Ic}.

1. Introduction – Fracture

Fracture = separation of a body into two or more pieces under stress. Steps : (1) crack initiation (nucleation) and (2) crack propagation. Fracture is the end of useful life (and often catastrophic) – bridges, Liberty ships (1943–46 : ≈ 1500 cracks, 12 broke in two in cold water), Comet airliners (fatigue at window corners), pipelines, pressure vessels, Titanic (brittle steel, high S/P, cold sea).

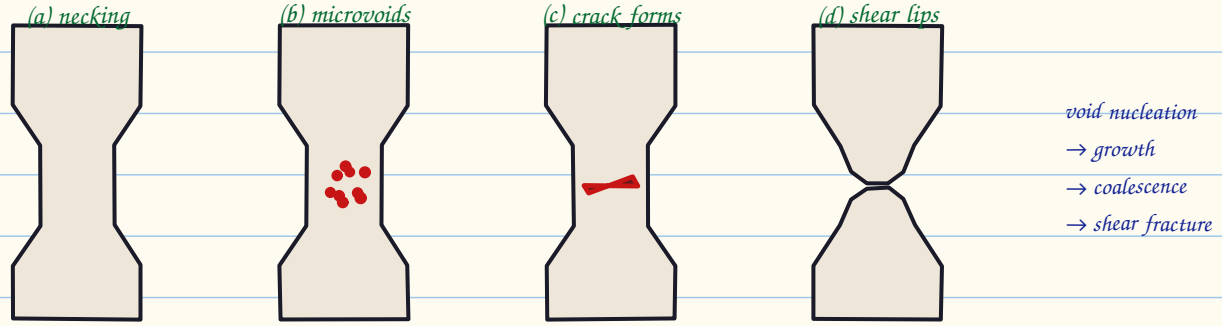
Classification of fracture

- **By ductility** : ductile (large plastic strain, slow stable crack, absorbs energy) / brittle (little plastic strain, fast unstable crack, flat surface) ; mixed (quasi-cleavage).
- **By crack path** : transgranular (through grains) / intergranular (along grain boundaries).
- **By loading** : tensile, shear, torsion, impact, **fatigue** (cyclic – Unit IV), **creep** (high T, time – Unit V), stress-corrosion, hydrogen-induced, thermal-shock fracture.



Feature	Ductile	Brittle
Plastic strain	large (> 5 % elongation)	negligible (< 1 %)
Crack speed	slow, needs increasing stress	fast, ≈ 0.4–0.5 × speed of sound
Energy absorbed	large	small
Warning	yes – visible necking	none
Surface	dull, fibrous, dimples	bright, granular / faceted, flat
Stress for propagation	increases with crack growth	may decrease (unstable)
Typical materials	Cu, Al, mild steel (> DBTT)	glass, ceramics, cast iron, BCC below DBTT

2. Ductile Fracture – Mechanism

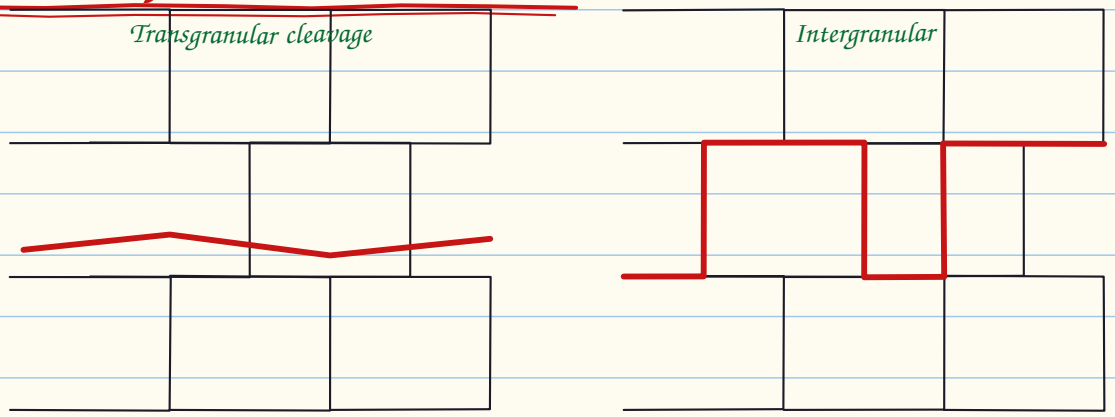


Cup-and-cone fracture of a round tensile bar (moderately ductile metals) :

- (a) After UTS a neck forms (plastic instability, $d\sigma = 0$) – local reduction of area, a **triaxial stress state** (hydrostatic tension) develops at the centre of the neck.
- (b) **Void (microcavity) nucleation** at inclusions (MnS, oxides), second-phase particles (carbides), by decohesion at the interface or particle cracking when local strain reaches a critical value.
- (c) **Void growth** (high triaxiality accelerates) and **coalescence** into an internal elliptical crack perpendicular to the axis at the centre of the neck ; ligaments between voids neck down (internal necking).
- (d) The crack grows outward until the outer rim, where the stress state is plane stress – it then **shears at ≈ 45°** (maximum shear stress) producing the **shear lip** (cone) ; the centre is flat, fibrous.

- **Fractography** : the centre shows equiaxed dimples (tensile), shear lip shows elongated (parabolic) dimples pointing to the origin ; dimple size depends on inclusion spacing – cleaner steels have deeper dimples and greater ductility.
- **Features** : high energy absorption, stable crack growth (needs increasing load), appears "fibrous". Very ductile metals (Au, Pb, Pt, soft Al) neck to a point (chisel/point rupture) ; ductile fracture in torsion is flat (shear), in thin sheets 45° slant fracture (single shear).
- Models : McClintock void growth ; Gurson–Tvergaard–Needleman (porous plasticity) ; critical void-volume fraction ≈ 0.15–0.25 for coalescence. **Ductility depends on** : inclusion content, stress triaxiality (notches reduce ductility), temperature, strain rate, texture.

3. Brittle Fracture – Mechanism



cracks cross grains on {100} planes

along weakened boundaries (temper embrittlement, H, S)

- Occurs with little or no plastic deformation, by rapid crack propagation perpendicular to the maximum tensile stress ; the crack runs at a high fraction of the speed of sound – failure without warning.
- **Cleavage** (transgranular) : separation along specific low-index crystal planes – {100} in BCC (Fe, W, Mo), {0001} in HCP (Zn, Mg), {111} in diamond cubic ; facets of cleavage show "river patterns" (steps merging in direction of crack growth) and **chevron / herringbone marks** on large fractures – the V's point back to the origin.
- **Intergranular** : crack path along grain boundaries – due to weak boundaries : grain-boundary films (e.g. cementite networks, S/P segregation), temper embrittlement (Sb, Sn, P segregation at 350–550 °C), hydrogen embrittlement,

liquid-metal embrittlement, stress-corrosion cracking, creep cavitation. Shiny, rock-candy appearance.

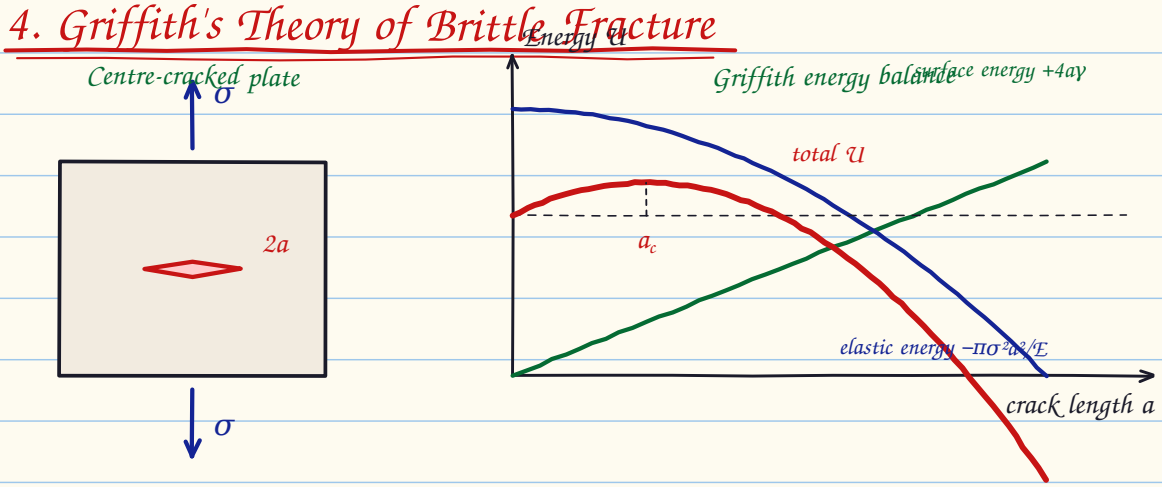
- **Mechanism of cleavage-crack nucleation** : dislocation pile-up of n dislocations on a slip plane against a grain boundary or a carbide ⇒ stress concentration $n\,\tau$; two interacting dislocations coalesce into a wedge crack nucleus (Cottrell : a/2[111] + a/2[1-1-1] → a[100]) ; or twin intersection ; or a cracked carbide particle (steels).

$$(\tau - \tau_i)\, k_y\, d^{1/2} \geq \beta\, G\, \gamma_{eff} \quad \text{(Stroh–Cottrell criterion)} \quad \sigma_f = \sigma_i + k_f\, d^{-1/2}$$

- Brittle fracture is favoured by : (a) triaxial stress (notch, thick section), (b) low temperature, (c) high strain rate (impact), (d) coarse grains, (e) BCC/HCP crystal structure, (f) hardened or irradiated steel. (The "three factors" of Orowan.)
- **Stress concentration** (Inglis) : $\sigma_{max} = \sigma[1 + 2\sqrt{(a/\rho)}] = K_t\,\sigma$ (a = semi-axis perpendicular to the load, ρ = tip radius) ; sharp cracks $\rho \to 0$ ⇒ stress → ∞ ⇒ need the energy criterion.

Ex 1. Elliptical hole a = 5 mm, ρ = 0.5 mm : $K_t = 1 + 2\sqrt{(5/0.5)} = 7.3$ ⇒ local stress 7.3 σ.

4. Griffith's Theory of Brittle Fracture



Observed fracture strength of brittle solids (glass ≈ 70–100 MPa) is far below the theoretical cohesive strength $\sigma_{th} \approx \sqrt{(E\gamma_s/a_0)} \approx E/10$ (≈ 8 GPa). **Griffith (1920)** : the solid contains pre-existing microcracks that raise the local stress ; a crack extends if the **elastic strain energy released** is at least equal to the **surface energy required** to create the new surfaces.

Energy balance for a through-crack of length 2a (unit thickness) in a large plate (plane

- Elastic energy released (stress relieved in a region $\approx \pi a^2$) : $\mathcal{U}_e = -\pi \sigma^2 a^2 / E$.
- Surface energy needed : $\mathcal{U}_s = 2\,(2a)\,\gamma_s = 4\,a\,\gamma_s$ (two faces).
- Total $\mathcal{U} = 4a\gamma_s - \pi\sigma^2 a^2/E$. The crack becomes unstable when $d\mathcal{U}/da = 0$:

$$4\gamma_s = 2\pi\sigma^2 a/E \Rightarrow \sigma_f = \sqrt{(\,2E\gamma_s / \pi\, a\,)} \text{ (plane strain : } E \to E/(1-\nu^2))$$

- Interpretation : for a given stress σ the critical crack length $a_c = 2E\gamma_s/(\pi\sigma^2)$; $\mathcal{U}$ reaches a maximum at a_c – shorter cracks close, longer cracks grow spontaneously (energy decreases). Fracture stress $\propto 1/\sqrt{a}$ – larger flaws ⇒ lower strength (strength scatter in ceramics ; size effect ; fibres).

Ex 2. Glass $E = 70$ GPa, $\gamma_s = 0.3$ J/m², flaw $a = 1$ µm : $\sigma_f = \sqrt{(2 \times 70\times10^9 \times 0.3/(\pi \times 10^{-6}))} = 116$ MPa ✓ (observed). For $a = 1$ nm (atomic flaw) $\sigma \approx 3.7$ GPa — close to the theoretical strength.

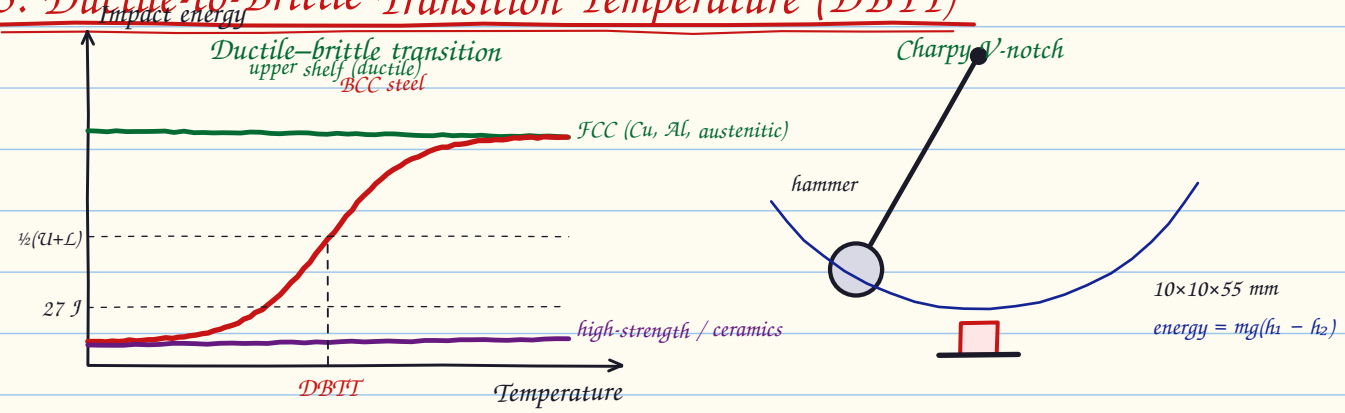
Orowan / Irwin modification for ductile metals

- Plastic work γ_p at the crack tip (10^3–$10^4 \times \gamma_s$ for steel) dominates :

$$\sigma_f = \sqrt{(\,2E\,(\gamma_s + \gamma_p)\,/\,\pi\, a\,)} \approx \sqrt{(\,2E\gamma_p\,/\,\pi\, a\,)}$$

- Griffith's theory is accurate for glass, ceramics, ice, silicon, polymers below T_g ; with the plastic-zone correction it works for metals (LEFM). **Limitations** : needs the pre-existing crack, ignores plastic zone & crack tip geometry.
- Hence strength of brittle materials is governed by flaw size ⇒ *surface finish, fire-polishing, tempering (compressive surface stress)* raise the strength of glass.

5. Ductile-to-Brittle Transition Temperature (DBTT)



Many materials, especially BCC metals (mild steel, Fe, Mo, W, Cr, Nb), fail by ductile fracture above a critical temperature and by brittle cleavage below it – the **ductile–brittle transition.** FCC metals (Cu, Al, Ni, austenitic steel) and very high-strength alloys do not show a sharp transition.

- **Cause** : the yield stress of BCC metals rises rapidly as temperature falls (high Peierls stress of screw dislocations) while the cleavage fracture stress is nearly temperature-independent. Below T_c the cleavage stress is reached before yielding ⇒ brittle ; Petch : $\beta\ T_c = \ln B - \ln(C\ d^{1/2})$ (finer grains lower T_c).
- **Charpy curve regions** : **upper shelf** – ductile (high energy, 100 % fibrous) ; **transition region** – mixed, large scatter ; **lower shelf** – brittle (≈ 5–20 J, 100 % cleavage).

Factors affecting DBTT

Factor	Effect	Direction
Carbon	≈ +14 °C per 0.1 % C ; pearlite (↑)	raises
Manganese	Mn : C > 3 ⇒ lowers (≈ −5 °C per 0.1 %), refines grain	lowers
Nickel	−10 to −20 °C per 1 % Ni ; 9 % Ni for LNG tanks (−196 °C)	lowers
P, S, O, N, Sb, Sn	segregation / inclusions / strain ageing	raise
Grain size	finer grain ⇒ lower DBTT (Petch)	finer lowers
Microstructure	tempered martensite (lowest) < bainite < pearlite+ferrite < coarse pearlite	—
Strain rate / temperature	high rate (impact) shifts curve to higher T	rate raises
Notch / triaxiality / thickness	sharp notch, thick plate, residual & welding stresses	raise
Cold work, hardening, irradiation	↑ yield, loss of toughness (neutron damage in reactor vessels +50–150 °C)	raise
Si, Al-killed practice	fully killed, normalised steel has lower DBTT	killing lowers

- Remedies : fine-grained killed Al steels, Ni additions, normalising/quench-tempering, controlled rolling (TMCP), avoid low-temperature

service or use FCC (austenitic stainless, Al, Cu), keep S & P low (< 0.01 %).

Determination of DBTT

- **Charpy V-notch impact test** (ASTM E23) : specimen 10 × 10 × 55 mm, 2 mm deep 45° notch, simply-supported, struck by a pendulum – energy absorbed = $mg(h_1 - h_2)$; tests at a series of temperatures (−196 → +100 °C, baths of liquid N_2, alcohol-dry ice, ice, hot water). **Izod** : cantilever mounting. Instrumented Charpy gives load–time curve.
- Criteria for DBTT (from the energy–temperature curve) : (1) mid-point of upper and lower shelf energy ; (2) fixed energy level – 20 J (15 ft-lb) for ships, 27 J (20 ft-lb) for structural steel ; (3) FATT – 50 % fibrous (shear) fracture appearance ; (4) lateral expansion 0.38 mm (15 mil) ; (5) NDT temperature.
- **Nil-ductility transition (NDT)** temperature – drop-weight test (Pellini, ASTM E208) : the highest T at which a specimen with a brittle weld bead breaks. Pellini's **fracture analysis diagram** : NDT, FTE (NDT + 33 °C : crack arrest), FTP (NDT + 67 °C : plastic overload) – used to choose service temperature (service T > FTE).

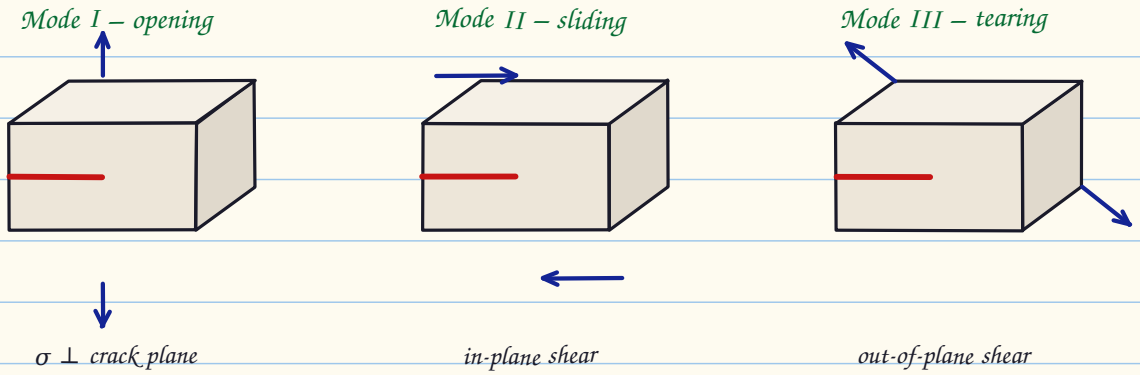
> Ex 3. Steel energy–temperature data : upper shelf 160 J, lower shelf 10 J ⇒ mid-point energy = 85 J ; reading the temperature at 85 J gives the DBTT ; the 27 J temperature is about 25 °C lower — specification may require 27 J at −20 °C.

6. Fracture Mechanics – Introduction

Conventional design uses $\sigma < \sigma_y/FS$; it ignores flaws. Fracture mechanics relates the applied stress, flaw size and material toughness – allows computing the safe stress, critical flaw size and inspection intervals (damage-tolerant design).

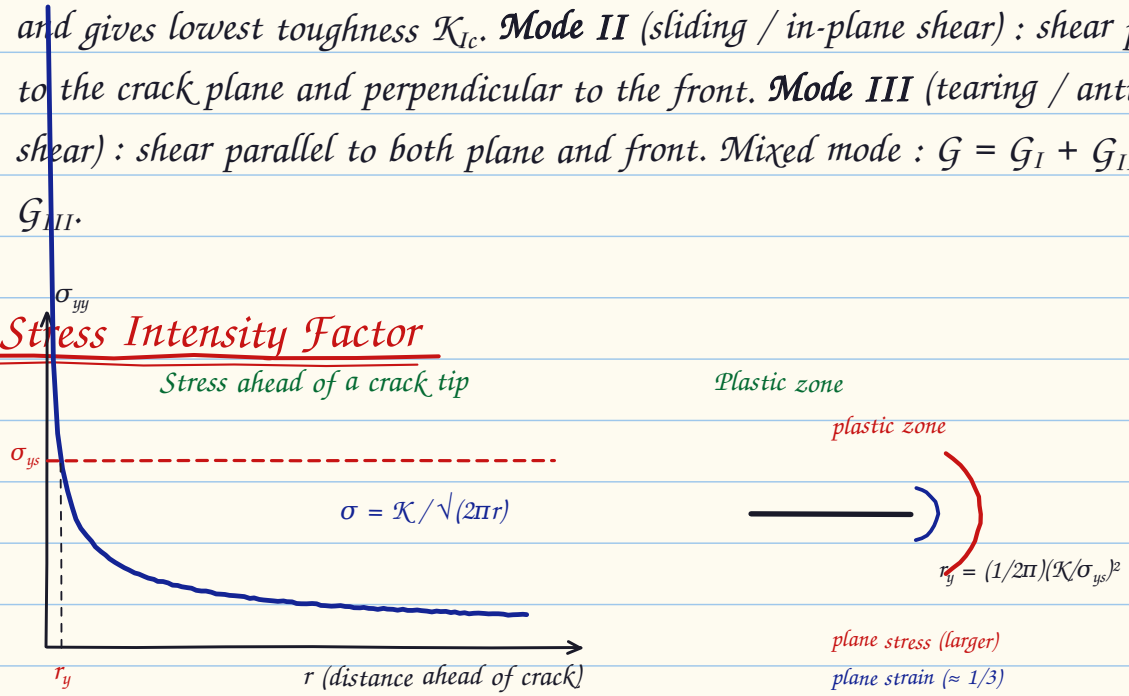
- **LEFM** (linear elastic fracture mechanics) : the plastic zone is small compared with crack length and specimen size. **EPFM** (elastic-plastic) : J-integral, CTOD for tough materials.

Modes of crack loading



- **Mode I** (opening) : tensile stress normal to the crack plane – the most important, and gives lowest toughness K_{Ic}. **Mode II** (sliding / in-plane shear) : shear parallel to the crack plane and perpendicular to the front. **Mode III** (tearing / anti-plane shear) : shear parallel to both plane and front. Mixed mode : $G = G_I + G_{II} + G_{III}$.

7. Stress Intensity Factor



Near the tip of a sharp crack ($r << a$) in a linear elastic body :

$$\sigma_{ij} = (K_I / \sqrt{(2\pi\ r)}) f_{ij}(\theta)\ K_I = Y \sigma \sqrt{(\pi\ a)}\ (MPa\sqrt{m})$$

- K_I is the **stress intensity factor** – a single parameter characterising the whole crack-tip field (stress $\propto K/\sqrt{r}$, singular) ; depends on applied stress σ, crack length a and geometry factor Y. Ahead of the crack on the plane $\theta = 0$: $\sigma_{yy} = K_I/\sqrt{(2\pi r)}$.
- Y : centre crack in an infinite plate 1.0 (a = half length) ; single edge crack 1.12 ($1.12\sqrt{\pi} = 1.99$) ; double edge crack ≈ 1.12 ; embedded penny crack $2/\pi$; surface half-circular crack 0.71 ; for finite width plates $Y = \sqrt{(\sec(\pi a/W))}$ (centre crack).
- **Fracture criterion** : unstable fracture occurs when $K_I = K_{Ic}$ (the plane-strain fracture toughness, a material property ; $MPa\sqrt{m}$). Safe if $K_I < K_{Ic}$.

$$\sigma_c = K_{Ic} / (Y\sqrt{(\pi\ a)})\ a_c = (1/\pi)\ [\ K_{Ic} / (Y\ \sigma)\ ]^2$$

Material	K_{Ic} (MPa$\sqrt{m}$)
Mild steel	140
4340 steel (σ_y 1500 MPa)	50
Maraging steel	90–110
Al 2024-T3 / 7075-T6	34 / 24–30
Ti-6Al-4V	55–115
Cast iron	6–20
Alumina / SiC	3–5 / 3–4
Glass	0.7
PMMA	1.0
Concrete	0.2–1.4

Ex 4. Plate with an edge crack, K_{Ic} = 50 MPa$\sqrt{m}$, σ = 300 MPa, Y = 1.12 : a_c = $(1/\pi)(50/(1.12\times300))^2$ = $(1/\pi)(0.1488)^2$ = 7.05 mm. If the largest crack found by inspection is a = 2 mm and K_{Ic} = 60 MPa$\sqrt{m}$: σ_c = $60/(1.12\sqrt{(\pi\times0.002)})$ = 676 MPa.

Plastic zone & small-scale yielding

- The elastic stress singularity is truncated by yielding : plastic-zone radius (Irwin)

$$r_y = (1/2\pi)(K_I/\sigma_{ys})^2 \text{ (plane stress)}\quad r_y = (1/6\pi)(K_I/\sigma_{ys})^2 \text{ (plane strain)}$$

- Effective crack length $a_{eff} = a + r_y$ (Irwin correction) – valid when $r_y < a/50$. Plane strain (thick plate) : triaxial constraint, small zone, flat fracture, lowest K_c ⇒ K_{Ic} ; plane stress (thin sheet) : larger zone, slant fracture, K_c higher.

Ex 5. K = 50 MPa$\sqrt{m}$, σ_{ys} = 1000 MPa : plane stress r_y = $(1/2\pi)(0.05)^2$ = 0.40 mm ; plane strain = 0.13 mm.

8. Strain Energy Release Rate

Energy released per unit crack-extension area (per unit thickness per unit crack advance) :

$$G = -(1/B)\ \partial U_e/\partial a = \pi\ \sigma^2\ a\ /\ E \text{ (plane stress)} ;\ G = \pi\ \sigma^2\ a\ (1-\nu^2)\ /\ E \text{ (plane strain)}$$

$$G_I = K_I^2 / E \quad E = E \text{ (plane stress)} ; E = E/(1 - \nu^2) \text{ (plane strain)}$$

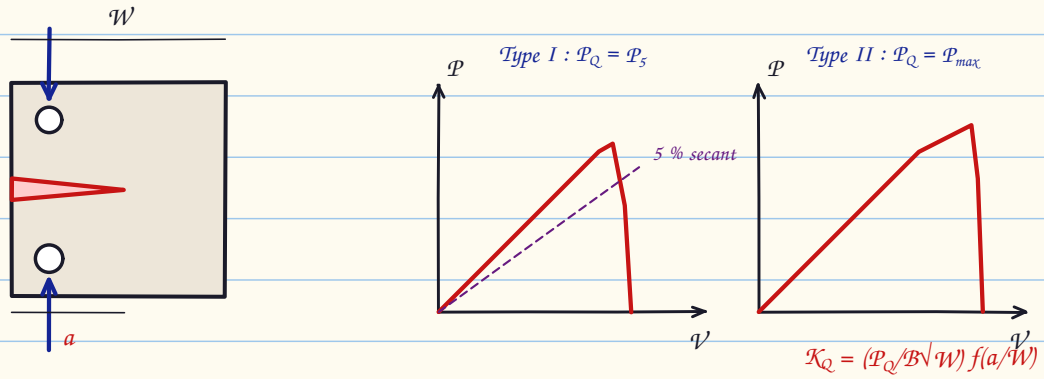
- Units J/m^2 (N/m). Crack growth when $G \geq G_c$ (critical strain energy release rate = energy needed to create new fracture surface $\approx 2\gamma_s + 2\gamma_p$) ; $G_{Ic} = K_{Ic}^2(1 - \nu^2)/E$. Equivalence of Griffith and Irwin : $K_I^2 = E\,G$.
- Griffith's relation from G : $G_c = 2\gamma_s \Rightarrow \sigma_f = \sqrt{(2E\gamma_s/\pi a)}$ (ideally brittle). ***Load control vs displacement control*** : in fixed-grip loading, G decreases with a (stable crack growth) ; constant load – G increases with a (unstable).
- ***R-curve*** : crack-growth resistance versus crack extension ; flat $R = G_c$ (plane strain), rising for thin sheets (plane stress) ; instability when $dG/da > dR/da$.

Ex 6. $K = 50\ MPa\sqrt{m}$, $E = 207\ GPa$: $G = K^2/E = (50\times10^6)^2/(207\times10^9) = 12.1\ kJ/m^2$ (plane stress) ; plane strain $\times(1 - 0.3^2) = 11.0\ kJ/m^2$.

- ***EPFM parameters*** : J-integral (energy flux, path independent) – $J = G$ in linear elasticity, J_{Ic} ; CTOD (crack-tip opening displacement) $\delta = K^2/(\sigma_{ys}E) = G/\sigma_{ys}$; for tough low-strength steels (ASTM E1820).

9. Fracture Toughness Testing – Determination of K_{Ic}

Compact tension (CT) Load–displacement records (E399)



ASTM E399 (plane-strain K_{Ic} testing) : specimens – compact tension (CT), single-edge notch bend (SENB, three-point), arc-shaped, disc compact.

- ***Specimen*** : CT with width $W = 2B$, $a/W = 0.45$–0.55 ; machined notch + fatigue pre-crack (sharp crack, $K_{max} \leq 0.6\ K_{Ic}$) of ≈ 1.3 mm.
- ***Test*** : load at slow rate (loading rate $dK/dt = 0.55$–$2.75\ MPa\sqrt{m}/s$) in a universal testing machine, record load P versus clip-gauge opening V (COD). Draw a secant line at 95 % of the initial slope (5 % secant) – intersection gives P_5 ; $P_Q = P_5$ if load before P_5 is lower (curve type I), or maximum load before P_5

(type II / III).

- Compute the conditional value

$$K_Q = (P_Q / (B\sqrt{W})) f(a/W)\, f(\alpha) = (2+\alpha)(0.886 + 4.64\alpha - 13.32\alpha^2 + 14.72\alpha^3 - 5.6\alpha^4) / (1-\alpha)^{3/2}$$

- **Validity (plane-strain) conditions** : $P_{max}/P_Q \le 1.10$ and

$$B,\ a,\ (W-a) \ge 2.5\,(K_Q/\sigma_{ys})^2$$

If valid, $K_Q = K_{Ic}$; else a thicker specimen is needed (K_c depends on thickness – plateau at $B \ge 2.5(K_{Ic}/\sigma_{ys})^2$).

Ex 7. CT : $B = 25$ mm, $W = 50$ mm, $a = 25$ mm ($\alpha = 0.5 \Rightarrow f = 9.66$), $P_Q = 20$ kN, $\sigma_{ys} = 500$ MPa : $K_Q = (20\times10^3/(0.025 \times \sqrt{0.05})) \times 9.66 = 3.58\ \text{MPa}\sqrt{\text{m}} \times 9.66 = 34.6\ \text{MPa}\sqrt{\text{m}}$. Check : $2.5(34.6/500)^2 = 0.0120$ m $= 12$ mm < 25 mm ✓ $\Rightarrow$ $K_{Ic} = 34.6\ \text{MPa}\sqrt{\text{m}}$. Required B for a 1000 MPa steel with $K_{Ic} = 50$: $B \ge 2.5(50/1000)^2 = 6.25$ mm.

- Other tests : SENB ($K = (P\,S/(B\,W^{3/2}))\, f(a/W)$) ; chevron-notch for ceramics (ASTM C1421) ; indentation fracture (ceramics) $K_c \approx 0.016\,(E/H)^{1/2}\,P/c^{3/2}$; J_{Ic} (ASTM E1820), CTOD (BS 7448), crack-arrest K_{Ia}.
- Uses : material selection (K_{Ic}/σ_{ys} ratio), critical flaw size, NDT inspection limits, leak-before-break design, remaining-life estimation. Increase of toughness : decrease strength, refine grain, clean steel, avoid embrittling phases, tempering ; metals with high K_{Ic}/σ_{ys} are fracture-tolerant.

10. Additional Solved Problems

P1. A tensile bar with a circumferential flaw $a = 1.5$ mm, $Y = 1.12$, $K_{Ic} = 40\ \text{MPa}\sqrt{\text{m}}$: $\sigma_c = 40/(1.12\sqrt{(\pi \times 0.0015)}) = 40/0.0769 = 520$ MPa.

P2. Ceramic ($E = 380$ GPa, $\gamma_s = 1\ \text{J/m}^2$, flaw $a = 20\ \mu$m) : $\sigma_f = \sqrt{(2\times380\times10^9\times1/(\pi\times20\times10^{-6}))} = 110$ MPa.

P3. G for $\sigma = 200$ MPa, $a = 10$ mm, $E = 200$ GPa : $G = \pi(200\times10^6)^2(0.01)/200\times10^9 = 6.3\ \text{kJ/m}^2 \Rightarrow K = \sqrt{(EG)} = 35.5\ \text{MPa}\sqrt{\text{m}}$ ($= \sigma\sqrt{(\pi a)} = 200 \times 0.1772\ \sqrt{}$).

P4. Which is safer, steel A (σ_{ys} 1500, K_{Ic} 60) or B (σ_{ys} 1000, K_{Ic} 100)? Critical a at $\sigma = 0.5\sigma_{ys}$ ($Y = 1.12$) : A : $(1/\pi)(60/(1.12\times750))^2 = 1.6$ mm ; B : $(1/\pi)(100/(1.12\times500))^2 = 10.2$ mm – steel B is far more fracture-tolerant.

Short answers – Quick Revision

- *Why is a brittle fracture dangerous?* No warning, rapid unstable growth, low energy absorption.
- *Why do FCC metals not show DBTT?* Yield stress is only weakly temperature dependent – dislocations stay mobile at low T.
- *Why is K_{Ic} measured under plane strain?* It is the lowest, conservative, thickness-independent toughness.
- *Relation between K and G?* $G = K^2/E$ ($E = E$ or $E/(1-\nu^2)$).
- *Use of Charpy test?* Qualitative ranking of notch toughness and DBTT, quality control ; not a design property (no direct link to K_{Ic}, but Barsom–Rolfe correlation).

11. Quick Formula Sheet – Unit III

Item	Formula
Stress concentration	$\sigma_{max} = \sigma(1 + 2\sqrt{(a/\rho)})$
Griffith	$\sigma_f = \sqrt{(2E\gamma_s/\pi a)}$; with plasticity $\sqrt{(2E(\gamma_s+\gamma_p)/\pi a)}$
Stress intensity	$K_I = Y\sigma\sqrt{(\pi a)}$; fracture if $K_I \geq K_{Ic}$
Critical crack	$a_c = (1/\pi)(K_{Ic}/Y\sigma)^2$
Plastic zone	$r_y = (1/2\pi)(K/\sigma_{ys})^2$; plane strain ÷3
Energy release rate	$G = K^2/E = \pi\sigma^2 a/E$
Validity (E399)	$B, a, W-a \geq 2.5(K_Q/\sigma_{ys})^2$; $P_{max}/P_Q \leq 1.1$

12. Important Questions

- Differentiate between ductile and brittle fracture; explain the cup-and-cone fracture mechanism.
- Explain the mechanism of cleavage fracture ; transgranular vs intergranular.
- Derive Griffith's criterion for brittle fracture ; state its modification for metals.

- What is DBTT? Discuss the factors affecting it and the methods of its determination (Charpy, NDT, FATT).
- Define modes of fracture, stress intensity factor, strain energy release rate ; derive the relation between G and K.
- Describe the determination of K_{Ic} (ASTM E399) with specimen, procedure and validity criteria.

UNIT – IV

Fatigue Behaviour & Testing

Subject : Mechanical Behaviour of Materials (23A03504c)

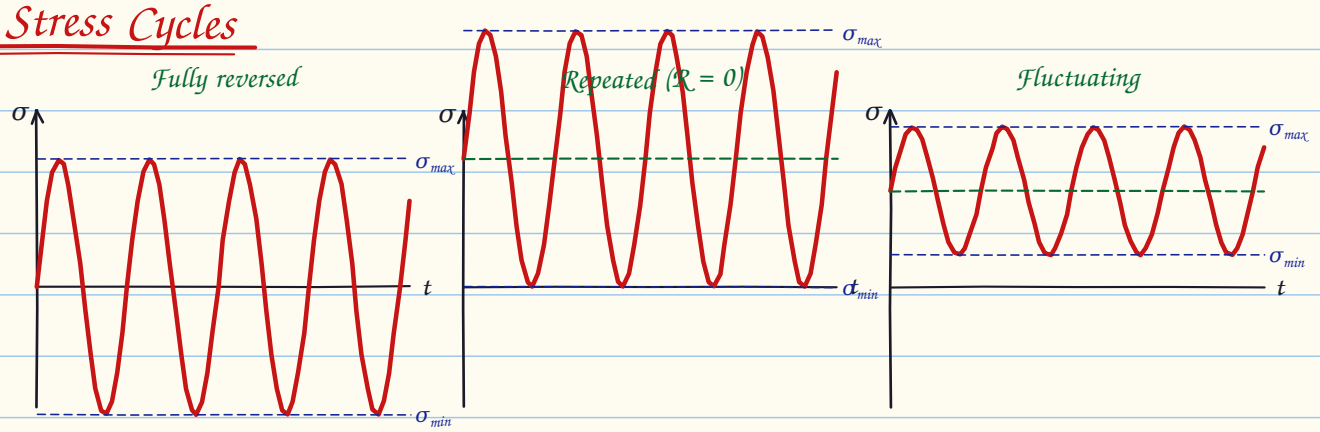
Syllabus: Stress cycles, S–N curves, effect of mean stress, factors affecting fatigue, structural changes accompanying fatigue, cumulative damage, HCF/LCF, thermo-mechanical fatigue, fracture mechanics applied to fatigue crack propagation (Paris law), fatigue testing machines.

1. Introduction – Fatigue

Fatigue = progressive, localised, permanent structural change in a material subjected to fluctuating stresses and strains that may culminate in cracks or complete fracture after a sufficient number of cycles – at stress levels well below the static UTS (often below yield).

- It accounts for ≈ 90 % of service failures of metallic parts (shafts, crankshafts, springs, aircraft wings, bridges, rails, gears, turbine blades, welded joints). Examples : Comet airliner (1954) window corners ; Versailles rail accident 1842 (axle) ; Alexander Kielland platform 1980.
- Characteristics : failure without warning ; crack starts at the surface (stress raisers) ; fracture surface has two zones – smooth fatigue zone with beach marks, and a rough final fast-fracture zone.
- Fatigue life $N_f = N_i$ (initiation) + N_p (propagation) – in HCF initiation dominates (≈ 90 %), in LCF propagation.

2. Stress Cycles



$\sigma_m = (\sigma_{max}+\sigma_{min})/2$; $\sigma_a = (\sigma_{max}-\sigma_{min})/2 = \Delta\sigma/2$; $R = \sigma_{min}/\sigma_{max}$; $A = \sigma_a/\sigma_m$

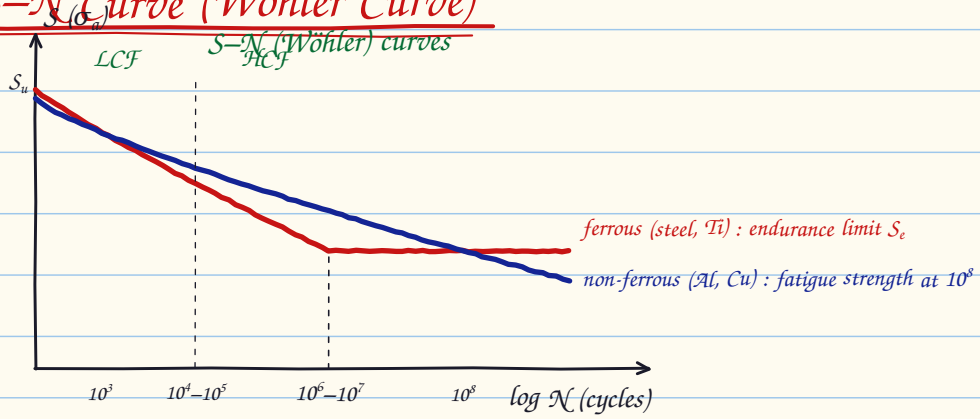
- **Stress range** $\Delta\sigma = \sigma_{max} - \sigma_{min}$. **Stress amplitude** $\sigma_a = \Delta\sigma/2$. **Mean stress** $\sigma_m = (\sigma_{max} + \sigma_{min})/2$. **Stress ratio** $R = \sigma_{min}/\sigma_{max}$; **amplitude ratio** $A = \sigma_a/\sigma_m = (1 - R)/(1 + R)$.

$$\sigma_{max} = \sigma_m + \sigma_a \quad \sigma_{min} = \sigma_m - \sigma_a$$

- Types : *fully reversed* ($\sigma_m = 0$, $R = -1$ – rotating shafts, R.R. Moore) ; *repeated / pulsating* ($\sigma_{min} = 0$, $R = 0$ – gear teeth, springs) ; *fluctuating* ($\sigma_m \neq 0$, general) ; random / spectrum (aircraft gust loads) ; block loading.

Ex 1. $\sigma_{max} = 300$ MPa, $\sigma_{min} = -100$ MPa : $\sigma_m = 100$ MPa ; $\sigma_a = 200$ MPa ; $\Delta\sigma = 400$ MPa ; $R = -100/300 = -0.33$.

3. S–N Curve (Wöhler Curve)



Plot of stress amplitude S (σ_a) versus the number of cycles to failure N (log scale), obtained by testing a set of identical specimens at different stress levels (usually $R = -1$).

- **Ferrous metals (steel, Ti)** : curve becomes horizontal at ≈ 10^6–10^7 cycles – the endurance (fatigue) limit S_e : the stress below which failure never occurs ; ≈ 0.4–0.5 S_u for steels ($S_e' = 0.5\ S_u$ for $S_u \le 1400$ MPa, capped at 700 MPa). Due to strain-ageing / locking of dislocations by interstitials (C, N).
- **Non-ferrous (Al, Cu, Mg) and aged steels** : no true limit ; the curve continues to fall – use fatigue strength at a specified life (10^7–5×10^8 cycles). $S_e \approx 0.4\ S_u$ at 5×10^8 for Al alloys.
- **Regions** : LCF ($N < 10^3$–10^5, plastic), HCF (10^5–10^7, elastic), infinite life. Basquin's law in the HCF region :

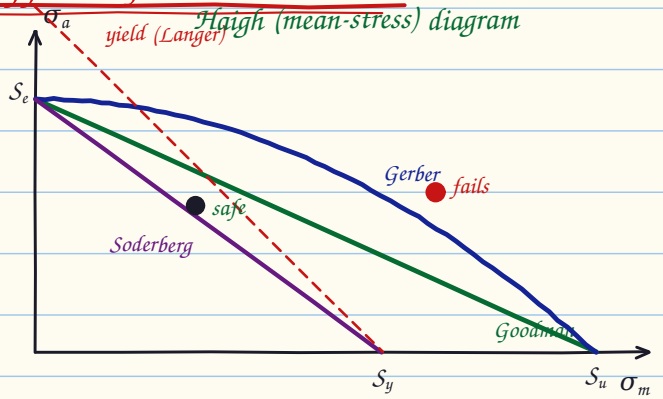
$$\sigma_a = \sigma_f' (2N_f)^b \quad (b = -0.05 \text{ to } -0.12 \ ; \ \sigma_f' \approx \text{true fracture strength})$$

- Scatter : ± factor 2–10 in life for the same stress – statistics (log-normal / Weibull) ; S–N–P curves ; **staircase method** to find S_e (≈ 15–30 specimens).

Ex 2. $\sigma_f' = 1000$ MPa, $b = -0.1$: for $\sigma_a = 300$ MPa : $2N = (0.3)^{-10} = 1.69\times10^5$ ⇒ $N = 8.5\times10^4$ cycles.

4. Effect of Mean Stress

Haigh (mean-stress) diagram



Goodman : $\sigma_a/S_e + \sigma_m/S_u = 1$
Gerber : $\sigma_a/S_e + (\sigma_m/S_u)^2 = 1$
Soderberg : $\sigma_a/S_e + \sigma_m/S_y = 1$
Compressive σ_m : little effect

A tensile mean stress reduces the allowable stress amplitude ; a compressive mean stress is beneficial. Empirical relations (Haigh diagram) – σ_a for infinite life at a given σ_m :

$$\text{Goodman} : \sigma_a/S_e + \sigma_m/S_u = 1 \quad \text{Gerber} : \sigma_a/S_e + (\sigma_m/S_u)^2 = 1 \quad \text{Soderberg} : \sigma_a/S_e + \sigma_m/S_y = 1$$

- **Goodman** – straight line, conservative for brittle materials (best fits tests) ; **Gerber** – parabola, fits ductile data ; **Soderberg** – most conservative, protects against yielding ; **Langer** – yield line $\sigma_a + \sigma_m = S_y$. Others : Morrow,

Smith–Watson–Topper (SWT : $\sigma_{max}\varepsilon_a E$).

- Factor of safety n : Goodman $1/n = \sigma_a/S_e + \sigma_m/S_u$. Compressive mean stress – assume $\sigma_m = 0$ (or small benefit).

> Ex 3. $S_e = 250$ MPa, $S_u = 600$ MPa, $S_y = 400$ MPa, $\sigma_m = 100$ MPa : Goodman $\sigma_a = 250(1 - 100/600) =$ 208 MPa ; Gerber $\sigma_a = 250(1 - 0.0278) =$ 243 MPa ; Soderberg $\sigma_a = 250(1 - 0.25) =$ 187.5 MPa.

> Ex 4. Component with $\sigma_{max} = 300$, $\sigma_{min} = 100$ MPa : $\sigma_m = 200$, $\sigma_a = 100$ MPa ⇒ Goodman $1/n = 100/250 + 200/600 = 0.733$ ⇒ $n =$ 1.36.

5. Factors Affecting Fatigue

The laboratory endurance limit S_e' is modified (Marin factors) for the real component :

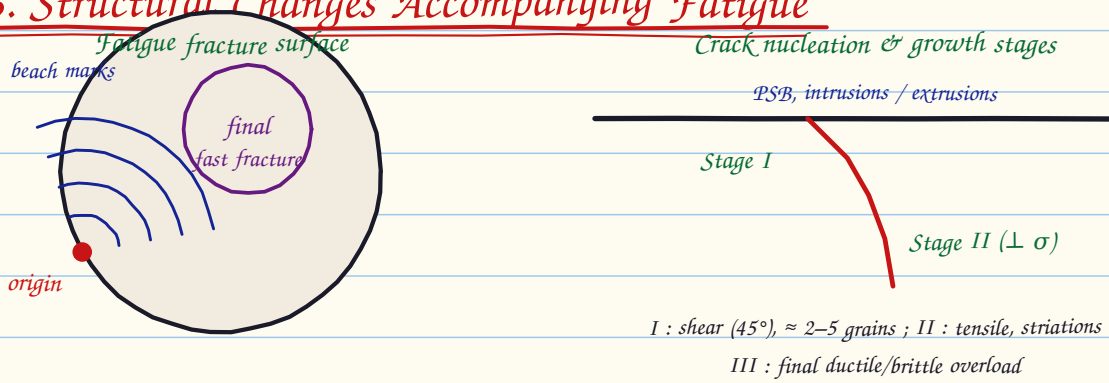
$S_e = k_a k_b k_c k_d k_e k_f S_e'$ $k_a = a S_u^{\,b}$ (surface) ; $k_b = 1.24\, d^{-0.107}$ (size, 2.8–51 mm) ; $k_c = 1$ (bending), 0.85 (axial), 0.59 (torsion) ; k_d (temperature) ; k_e (reliability 0.897 for 90 %, 0.814 for 99 %) ; k_f miscellaneous

- **1. Stress concentration** (holes, notches, keyways, threads, fillets, sharp corners, welds) : fatigue notch factor $K_f = 1 + q\,(K_t - 1)$ (q = notch sensitivity 0–1, rises with notch radius and material strength) ; S_e (notched) $= S_e/K_f$. Remedy : large fillet radii, smooth transitions, stress-relief grooves, avoid abrupt section changes.
- **2. Surface finish** : polished surfaces best ; machined, hot-rolled, forged reduce S_e (scratches = micro-notches, decarburisation). k_a (machined) $= 4.51\, S_u^{-0.265}$.
- **3. Size effect** : larger section ⇒ larger stressed volume ⇒ more flaws, steeper gradient ⇒ lower S_e.
- **4. Residual stresses** : compressive surface stresses (shot peening, cold rolling of fillets, burnishing, carburising, nitriding, induction hardening, autofrettage) increase fatigue life 20–100 % (retard crack initiation) ; tensile residual stresses (welding, grinding, quenching) reduce it. Removed by high T / overload.
- **5. Environment** : corrosion fatigue – synergy of cyclic stress and corrosive medium ⇒ no endurance limit, S–N curve keeps falling ; fretting fatigue at contact surfaces (press fits, bolts) – loses 30–70 % ; hydrogen ; high T (creep–fatigue).
- **6. Metallurgical** : grain size (fine grain ↑ S_e), inclusions/porosity (casting defects, ↓), banding, anisotropy (transverse < longitudinal), decarburised layer, microstructure (tempered martensite best), hardness ($S_e \approx 1.7$–1.8 HB up to 400 HB). **7. Temperature, frequency, overloads, mean stress, type of loading, surface**

treatments (chromium plating reduces S_e), coatings.

Ex 5. Machined steel shaft $d = 20$ mm, $S_u = 600$ MPa, bending : $S_e' = 300$ MPa ; $k_a = 4.51(600)^{-0.265} = 0.83$; $k_b = 1.24(20)^{-0.107} = 0.90$; $S_e = 300 \times 0.83 \times 0.90 = 224$ MPa. With a shoulder fillet $K_t = 2.5$, $q = 0.8$: $K_f = 1 + 0.8 \times 1.5 = 2.2$ ⇒ allowable amplitude $224/2.2 \approx 102$ MPa.

6. Structural Changes Accompanying Fatigue



Stages of fatigue failure

- **Stage I – crack initiation** : repeated cyclic slip on the most favourably oriented systems forms persistent slip bands (PSBs) (concentrated slip, dislocation ladder structure in Cu : veins and walls) ; slip steps at the surface form **extrusions** (ribbons pushed out) and **intrusions** (micro-notches) – the nucleus of the crack, at grain boundaries, inclusions, notches, in about the first 10 % of life. Crack grows along the slip plane (≈ 45° to σ) through 2–5 grains – **Stage I growth** (shear mode).
- **Stage II – crack propagation** : the crack turns normal to the maximum tensile stress and advances by blunting and re-sharpening at the tip each cycle, leaving **striations** (one per cycle, spacing ≈ da/dN, 0.01–1 μm, visible in the SEM) ; larger **beach (clamshell) marks** mark changes in loading or interruptions, seen by naked eye, concentric around the origin.
- **Stage III – final fracture** : when the remaining cross-section can no longer carry the load ($K_{max} = K_c$) ⇒ sudden ductile (tear) or brittle fracture. Relative area of the fast-fracture zone ⇒ indicates the stress level (small zone ⇒ low stress, large fatigue zone).

Cyclic deformation

- **Cyclic hardening** (annealed metals – dislocation multiplication) or **cyclic softening** (cold-worked/hard metals – dislocations rearrange) until a saturated hysteresis loop is reached ; the **cyclic stress–strain curve** (locus of loop tips) differs from the monotonic curve : cyclic hardening if $n' > n$. Indicator : if $S_u/S_y > 1.4$ hardens, < 1.2 softens.
- **Bauschinger effect,** hysteresis-loop energy per cycle ≈ plastic damage ; at high temperature cracks become intergranular (creep–fatigue).
- **Fatigue crack closure** (Elber) : plasticity-induced residual wake closes the crack before zero load ⇒ effective $\Delta K_{eff} = K_{max} - K_{op}$.

7. Cumulative Damage

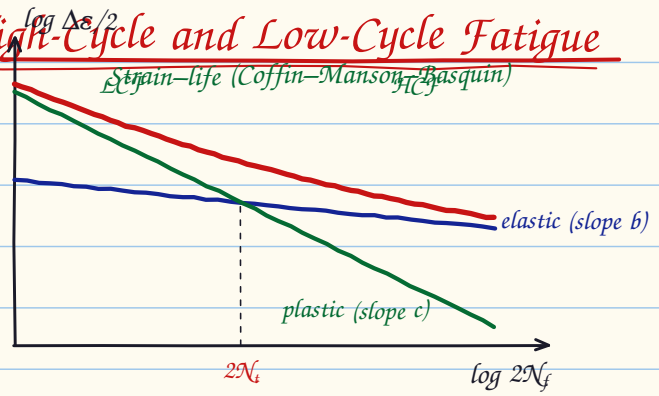
Real components see variable amplitude loads. **Palmgren–Miner linear damage rule** : each cycle at stress level i causes damage $1/N_i$; failure when the sum reaches 1 :

$$D = \Sigma \left(n_i / N_i \right) = 1$$

- n_i = cycles applied at amplitude σ_i, N_i = life at that amplitude from the S–N curve. Remaining life fraction = $1 - D$.
- **Limitations** : ignores load sequence (high-to-low gives $\Sigma < 1$: failure earlier ; low-to-high gives $\Sigma > 1$), neglects interaction effects, notch plasticity, residual stresses, stress below the endurance limit (they do cause damage after cracks start) ; real $\Sigma\ n/N = 0.3–3$. Improved : **Marco–Starkey** ($D = \Sigma\ (n/N)^x$), **Manson's double linear damage rule, Corten–Dolan,** damage curve approaches ; **rainflow counting** (ASTM E1049) to extract cycles from a random history.

Ex 6. A part is run 2×10^4 cycles at σ_1 ($N_1 = 10^5$), 5×10^4 at σ_2 ($N_2 = 5\times10^5$) and 2×10^5 at σ_3 ($N_3 = 10^6$) : $D = 0.2 + 0.1 + 0.2 = 0.5$ ⇒ remaining life 50 % ; if the remainder is at σ_1 : further $n = 0.5 \times 10^5 = 5\times10^4$ cycles.

8. High-Cycle and Low-Cycle Fatigue



$\Delta\varepsilon/2 = (\sigma_f'/E)(2N_f)^b + \varepsilon_f'(2N_f)^c$

$b \approx -0.05$ to -0.12

$c \approx -0.5$ to -0.7

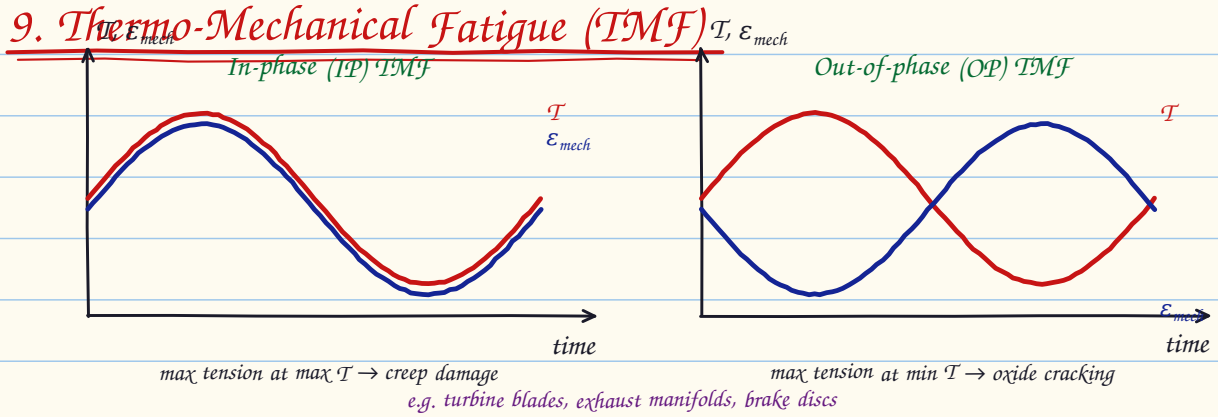
- **HCF** : $N_f > 10^4$–10^5 ; nominal stresses elastic ; stress-controlled – S–N approach ; fatigue life governed by crack initiation ; examples : shafts, springs, bridges, machine parts.
- **LCF** : $N_f < 10^4$–10^5 ; large plastic strain each cycle (local yielding at notches) ; strain-controlled – ε–N approach ; hysteresis loops visible ; crack propagation dominates ; examples : pressure vessel start-up cycles, turbine disc, thermal cycling, earthquake loading.
- **Coffin–Manson** (plastic strain) : $\Delta\varepsilon_p/2 = \varepsilon_f'\ (2N_f)^c$ ($c \approx -0.5$ to -0.7, $\varepsilon_f' \approx$ true fracture ductility). Elastic part (Basquin) : $\Delta\varepsilon_e/2 = (\sigma_f'/E)(2N_f)^b$. **Total strain–life** (Manson–Coffin–Basquin–Morrow) :

$$\Delta\varepsilon/2 = (\sigma_f'/E)(2N_f)^b + \varepsilon_f'\ (2N_f)^c$$

- **Transition life** $2N_t$ where plastic and elastic strain amplitudes are equal ; ductile materials have large $2N_t$ (strain-controlled design), strong brittle materials small $2N_t$ (stress-controlled). **Universal slopes** (Manson) : $\Delta\varepsilon = 3.5\ (S_u/E)\ N^{-0.12} + D^{0.6}\ N^{-0.6}$.

Ex 7. $\varepsilon_f' = 0.5$, $c = -0.6$: a plastic strain amplitude $\Delta\varepsilon_p/2 = 0.005 \Rightarrow (2N)^{-0.6} = 0.01 \Rightarrow 2N = 100^{1/0.6} = 2154 \Rightarrow N = \approx$ 1080 cycles.

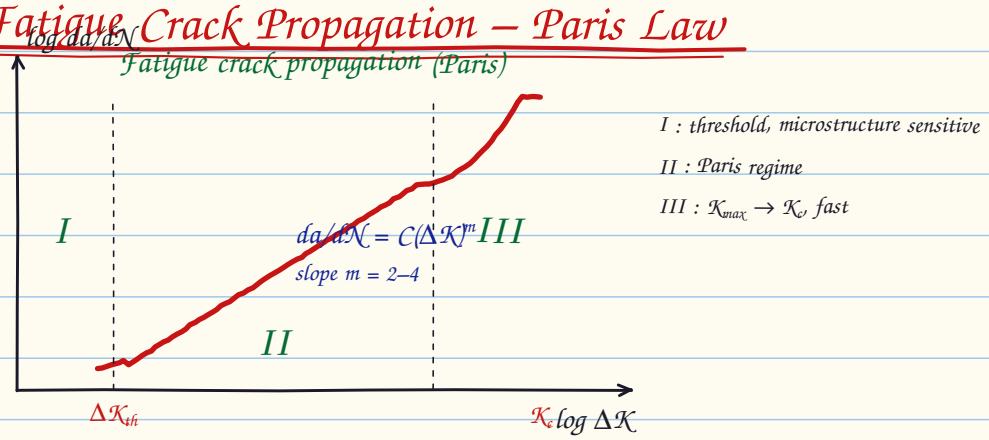
9. Thermo-Mechanical Fatigue (TMF)



TMF : fatigue under simultaneously varying temperature and mechanical strain (not isothermal) – thermal expansion is constrained by the neighbouring material, giving cyclic thermal strain $\varepsilon_{th} = \alpha\,\Delta T$. Total strain $\varepsilon_{mech} = \varepsilon_{total} - \alpha(T - T_0)$.

- Thermal stress for a fully constrained part : $\sigma = E\,\alpha\,\Delta T$ — e.g. steel E = 210 GPa, $\alpha = 12\times10^{-6}/°C$, ΔT = 300 °C ⇒ σ = 756 MPa.
- **Phasing** : in-phase (IP – max tensile strain at max T ⇒ creep damage, ductile) ; out-of-phase (OP – max tension at min T ⇒ oxide-assisted cracking at the surface, more damaging for Ni superalloys, Al). Also diamond cycles, clockwise/counter-clockwise.
- Damage mechanisms : fatigue + creep + oxidation (environmental) interaction ; life models – Neu–Sehitoglu, strain-range partitioning (PP, PC, CP, CC), damage-accumulation, Larson–Miller. Examples : turbine blades/vanes, combustor liners, exhaust manifolds, brake discs, solder joints, nuclear plant piping (thermal shock). Testing : servo-hydraulic machine with induction/resistance heating, extensometer, controlled temperature–strain waveform.
- Reduction : lower thermal gradients, compliant design, materials with low α/E and high conductivity (reduce $E\alpha/k$), thermal barrier coatings.

10. Fatigue Crack Propagation – Paris Law



Fracture mechanics treats the crack growth rate as a function of the stress intensity range $\Delta K = K_{max} - K_{min} = Y \Delta\sigma \sqrt{(\pi a)}$ ($\Delta\sigma$ = tensile range ; compressive part ignored).

- Three regions of the log(da/dN)–log ΔK curve : **I** – below the threshold ΔK_{th} (≈ 2–5 MPa√m for steels ; no growth ; microstructure & R sensitive) ; **II** – linear (Paris) region ; **III** – rapid growth as $K_{max} \to K_c$ (or K_{Ic}) : unstable.

$da/dN = C\,(\Delta K)^m$ (Paris–Erdogan 1963) ; m = 2–4 (steels ≈ 3, Al ≈ 3–4), C from tests (units m/cycle with ΔK in MPa√m)

- **Forman** equation (includes R, region III) : $da/dN = C(\Delta K)^m / [(1 - R)K_c - \Delta K]$; **Walker** : $da/dN = C[\Delta K/(1 - R)^{1-\gamma}]^m$.
- **Fatigue life by integration** from the initial flaw a_i to the critical size a_f :

$N_p = \int_{a_i}^{a_f} da / [\, C\,(Y \Delta\sigma \sqrt{(\pi a)})^m \,] = [\, a_i^{1-m/2} - a_f^{1-m/2} \,] / [\, (m/2 - 1)\, C\, Y^m\, \Delta\sigma^m\, \pi^{m/2} \,]$ ($m \neq 2$, Y const.)

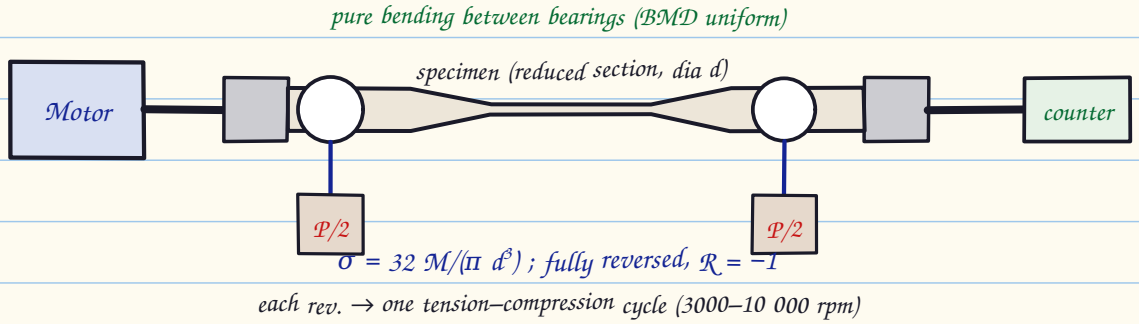
with $a_f = (1/\pi)[K_{Ic}/(Y\,\sigma_{max})]^2$.

Ex 8. $m = 3$, $C = 10^{-11}$ (m/cycle, MPa√m), $Y = 1.12$, $\Delta\sigma = 100$ MPa ($R = 0 \Rightarrow \sigma_{max} = 100$), $K_c = 60$ MPa√m, $a_i = 1$ mm : $a_f = (1/\pi)(60/112)^2 = 91$ mm ; $N_p = [31.6 - 3.31]/[0.5 \times 10^{-11} \times 1.405 \times 10^6 \times 5.568] = 28.3/(3.91\times10^{-5}) = 7.2\times10^5$ cycles. (Most of the life is spent while the crack is small – inspection must detect a_i.)

- Uses : damage-tolerant design (aircraft, rail, pipelines) – set inspection intervals ; retardation by overloads (plastic zone), acceleration by under-loads ; short cracks grow faster than predicted ; effect of mean stress via R. **Increasing fatigue crack growth resistance** : higher K_{Ic}, compressive residual stresses, fine grains (near threshold), avoid corrosive environment.

11. Fatigue Testing Machines

R.R. Moore rotating-beam fatigue machine



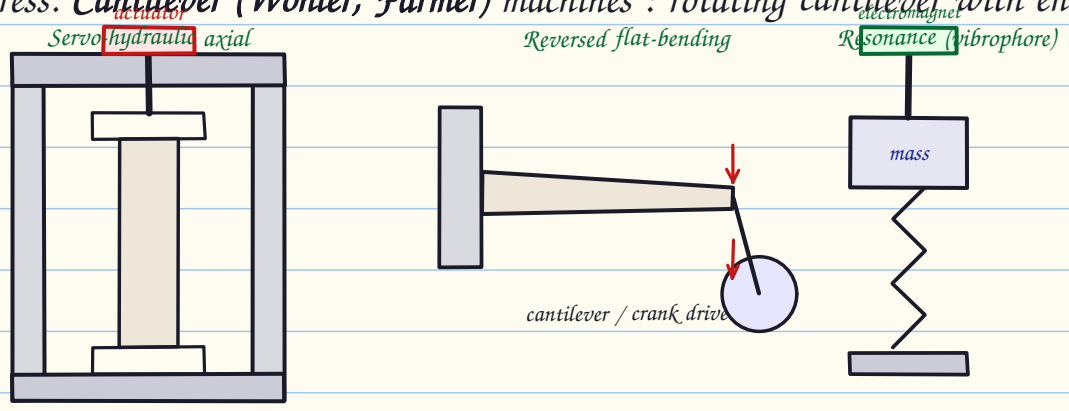
(a) R.R. Moore rotating-beam machine (pure bending)

- A motor-driven, hour-glass specimen is held in two bearings and loaded by weights so that the middle zone sees **pure bending (constant bending moment)** ; as the specimen rotates each fibre experiences tension–compression : fully reversed stress $R = -1$; speed 3000–10 000 rpm ; counter records cycles ; stops automatically at fracture.

$$\sigma = 32\,M / (\pi\, d^3)\ (M = \text{bending moment},\ d = \text{diameter at minimum section})$$

Ex 9. $d = 8$ mm, target $\sigma = 300$ MPa $\Rightarrow M = \sigma\pi d^3/32 = 300\times10^6 \times \pi \times (0.008)^3/32 = 15.1$ N·m (e.g. 151 N at an arm of 0.1 m).

- Cheap, simple, high speed ; limited to reversed bending, small specimens, no mean stress. **Cantilever (Wöhler, Farmer)** machines : rotating cantilever with end load.



(b) Axial (push–pull) servo-hydraulic machine

- Hydraulic actuator + servo-valve + load cell + extensometer under closed-loop digital control (load, strain or displacement waveform – sine, triangle, random, block) ; frequency 0.1–100 Hz ; R any value, mean stress, LCF (strain-controlled),

crack-growth (K-controlled, ASTM E647), TMF with furnace ; Instron, MTS.

(c) Other machines

- **Reversed flat-bending** (sheet – cantilever bent by a crank/eccentric) ; **torsional fatigue** machines ; **resonance (vibrophore) machines** : electro-magnetic or electromechanical excitation at the natural frequency (50–250 Hz) of a spring–mass system – low energy cost, fast, ideal for HCF ; **ultrasonic fatigue** (20 kHz – 10^9–10^{10} cycles in a few days, very high cycle fatigue) ; **rotating-bending with combined axial loads** ; **full-scale component test rigs** (axles, wheels, wings).
- Standards : ASTM E466 (force-controlled axial), E606 (strain-controlled), E647 (crack growth), E739 (statistical analysis), ISO 1099, ISO 12106.
- **Procedure for S–N curve** : 8–12 identical polished specimens (surface finish < 0.2 µm) ; first at $\sigma \approx 0.7\,S_u$, then reduce in steps ; run-out at 10^7 (steel) ; plot log N vs S, draw best fit curve ; for endurance limit use staircase (up-down) method.

12. Additional Solved Problems

P1. A steel shaft sees $\sigma_a = 220$ MPa fully reversed ; $S_u = 700$ MPa, $S_e' = 350$ MPa, $k_a = 0.8$, $k_b = 0.85 \Rightarrow S_e = 350 \times 0.8 \times 0.85 = 238$ MPa $\Rightarrow$ factor of safety $n = 238/220 = 1.08$ – inadequate for design.

P2. Paris-law crack grows from 2 mm to 12 mm at $\Delta\sigma = 80$ MPa, $Y = 1.0$, $m = 2.5$, $C = 5\times10^{-12}$: $a_i^{-0.25} = 4.73$, $a_f^{-0.25} = 3.02 \Rightarrow$ difference 1.71 ; denominator = $0.25 \times 5\times10^{-12} \times 80^{2.5} (= 57\,243) \times \pi^{1.25} (= 4.19) = 3.0\times10^{-7} \Rightarrow N = 1.71/3.0\times10^{-7} = 5.7\times10^6$ cycles.

P3. Goodman : $S_u = 500$, $S_e = 220$ MPa : at $\sigma_m = 150$ MPa, $\sigma_{a,allowed} = 220(1 - 0.3) = 154$ MPa.

P4. $R = 0.1$, $\sigma_{max} = 200$ MPa : $\sigma_{min} = 20 \Rightarrow \sigma_a = 90$ MPa ; $\sigma_m = 110$ MPa ; $A = 0.82$.

Short answers – Quick Revision

- *What is the endurance limit?* Stress amplitude below which a ferrous material has infinite life ($\approx 0.5\,S_u$) ; absent in Al, Cu and corrosive environments.
- *Why does fatigue crack start at the surface?* Maximum stress, free surface slip, stress raisers, environment.
- *What do striations represent?* One per load cycle in Stage II ; spacing ≈ da/dN.

- *Why compressive residual stress is beneficial?* It lowers the effective mean/tensile stress that opens the crack.
- *Difference HCF and LCF?* Elastic stress-controlled > 10^4–10^5 cycles vs plastic strain-controlled < 10^4.

13. Quick Formula Sheet – Unit IV

Item	Formula
Stress cycle	$\sigma_a = (\sigma_{max} - \sigma_{min})/2$; $R = \sigma_{min}/\sigma_{max}$
Basquin	$\sigma_a = \sigma_f' (2N)^b$
Goodman	$\sigma_a/S_e + \sigma_m/S_u = 1$
Marin	$S_e = k_a k_b k_c k_d k_e S_e'$
Notch	$K_f = 1 + q(K_t - 1)$
Miner	$\Sigma\ n_i/N_i = 1$
Strain–life	$\Delta\varepsilon/2 = (\sigma_f'/E)(2N)^b + \varepsilon_f' (2N)^c$
Paris	$da/dN = C(\Delta K)^m$; $\Delta K = Y\Delta\sigma\sqrt{(\pi a)}$
Rotating beam	$\sigma = 32M/\pi d^3$

14. Important Questions

- Define stress cycle terms ; explain S–N curve, endurance limit and fatigue strength. Why do ferrous and non-ferrous metals differ?
- Explain the effect of mean stress (Goodman, Gerber, Soderberg) with a Haigh diagram.
- List and explain the factors affecting fatigue strength ; role of residual stress and surface treatment.
- Describe the stages of fatigue failure and structural changes (PSBs, intrusions, striations, beach marks).
- Explain Miner's rule of cumulative damage with limitations. Differentiate HCF and LCF ; Coffin–Manson relation.
- What is thermo-mechanical fatigue? Explain IP and OP cycles.
- Derive/Explain Paris law and the three regions of crack growth ; compute fatigue life by integration.

- Describe the R.R. Moore rotating-beam machine and axial servo-hydraulic machines.

UNIT – V

Creep Behaviour & Testing

Subject : Mechanical Behaviour of Materials (23A03504c)

> Syllabus: Creep curve and its stages, structural changes during creep, creep mechanisms, metallurgical factors affecting creep, high-temperature alloys, stress-rupture testing, creep testing machines.

1. Introduction – Creep

Creep = time-dependent, permanent deformation of a material under a constant stress (or load) that is usually below the yield strength, occurring at high temperature – generally $T > 0.4\ T_m$ (homologous temperature T/T_m, T in kelvin).

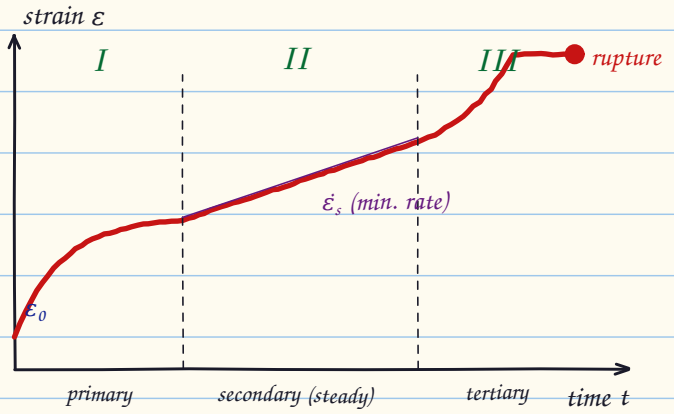
- Examples : turbine blades and discs, boiler/steam pipes and superheater tubes, nuclear fuel cladding, furnace parts, jet engines, rocket nozzles, heat-exchangers, solder joints, lead pipes and roofs at room temperature (Pb T_m = 600 K ⇒ RT = 0.49 T_m), tungsten lamp filaments (sagging), glacier flow.
- Critical temperature for creep (0.4 T_m) : Fe 451 °C ; Ni 418 °C ; Al ≈ 100 °C (0.4×933 = 373 K) ; Cu 270 °C ; Ti 530 °C ; W 1200 °C ; Pb, Sn ≈ room T ; polymers near RT. Tungsten at 1000 °C (0.35 T_m) hardly creeps, while Al at 150 °C does.
- Failure modes in service : excessive deformation (loss of clearance – blade tip rubbing), creep rupture (fracture after long time), stress relaxation of bolts/springs (creep under constant strain), creep-fatigue interaction. Design life 10^5 h (≈ 11.4 years) for power plant.

Metal	T_m (K)	0.4 T_m (K)	0.4 T_m (°C)
Pb	600	240	–33
Al	933	373	100
Cu	1356	542	269
Fe	1811	724	451
Ni	1728	691	418

Metal	T_m (K)	0.4 T_m (K)	0.4 T_m (°C)
Ti	1941	776	503
W	3683	1473	1200

2. The Creep Curve

Creep curve (constant load, T)



$\varepsilon = \varepsilon_0 + \varepsilon_t(1 - e^{-rt}) + \dot{\varepsilon}_s\, t$ (Garofalo)

I : decreasing rate (hardening)
II : hardening = recovery
III : necking, voids, rupture

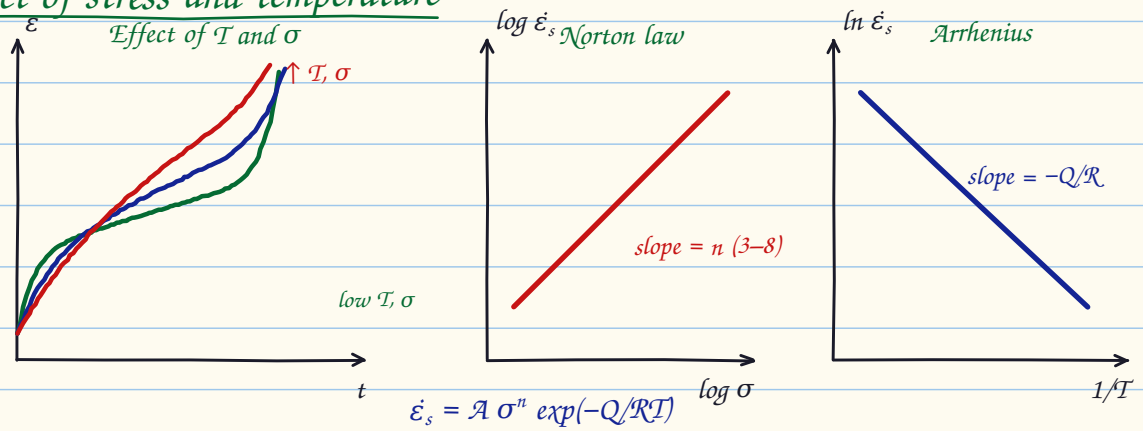
Strain ε is plotted against time t at constant temperature and constant load (stress). Three stages after the instantaneous strain :

- **Instantaneous (initial) strain ε_0** : elastic (+ some plastic) strain on loading.
- **Stage I – primary (transient) creep** : creep rate $\dot{\varepsilon} = d\varepsilon/dt$ decreases with time ; strain hardening dominates – dislocations multiply and tangle, the material becomes more resistant.
- **Stage II – secondary (steady-state, viscous) creep** : approximately constant minimum creep rate $\dot{\varepsilon}_s$; a dynamic balance of strain hardening and recovery (softening) – dislocations climb, annihilate, form subgrains. Longest stage, used in design (creep rate × life).
- **Stage III – tertiary creep** : rate accelerates leading to rupture ; caused by necking (area reduction at constant load increases true stress), internal cavities / grain-boundary cracks, microstructural degradation (coarsening of precipitates, over-ageing, recrystallisation) and oxidation.

$\varepsilon = \varepsilon_0 + \varepsilon_t(1 - e^{-rt}) + \dot{\varepsilon}_s\, t$ (Garofalo ; transient + steady) ; low-T logarithmic creep $\varepsilon = \alpha\, \ln(1 + \beta t)$

- **Andrade** : $\varepsilon = \varepsilon_0(1 + \beta\, t^{1/3})\, e^{Kt}$. Constant-stress tests (cam compensation) show a longer secondary stage and no tertiary stage from necking (only structural) ; **constant-load** tests are standard in industry.

Effect of stress and temperature



$\dot{\varepsilon}_s = A\,\sigma^n \exp(-Q/RT)$

- Higher T or σ ⇒ larger ε_0 and $\dot{\varepsilon}_s$, shorter primary stage, earlier tertiary stage, shorter rupture life t_r ; very low T/σ ⇒ only primary creep (strain decays, creep stops).
- **Steady-state creep rate** :

$$\dot{\varepsilon}_s = A\,\sigma^n \exp(-Q_c/RT)$$ (Norton power law × Arrhenius ; Mukherjee–Bird–Dorn)

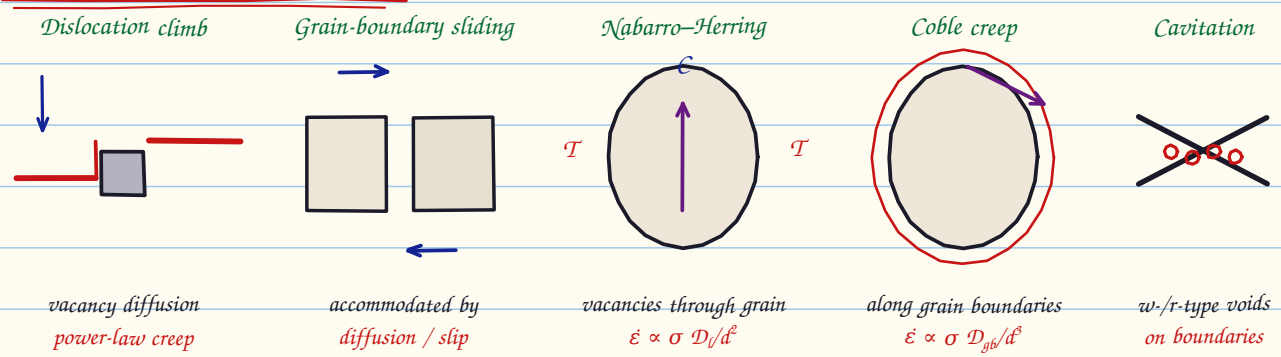
A = constant (structure) ; n = stress exponent (3–8 for dislocation (climb) creep ; 1 for diffusional creep) ; Q_c = activation energy for creep ≈ Q_{SD} (self-diffusion : Al 142, Cu 200, Fe 280, Ni 285 kJ/mol) ; R = 8.314 J/mol·K.

- **Monkman–Grant relation** : $\dot{\varepsilon}_s \cdot t_r$ = constant (≈ 0.01–0.1) ⇒ rupture life inversely proportional to the minimum creep rate. **Creep ductility** ε_r = strain at rupture (5–40 %).

Ex 1. (n) $\dot{\varepsilon} = 10^{-8}\ h^{-1}$ at 100 MPa and 10^{-6} at 200 MPa (same T) : n = log(100)/log2 = 6.6. At 120 MPa : $\dot{\varepsilon} = 10^{-7}(1.2)^{6.6} = 3.4\times10^{-7}\ h^{-1}$ ⇒ time to 1 % creep = $0.01/3.4\times10^{-7} \approx 3\times10^4$ h.

Ex 2. (Q) $\dot{\varepsilon} = 10^{-9}\ h^{-1}$ at 800 K and 10^{-8} at 850 K (same σ) : $Q = R\,\ln10 /(1/800 - 1/850) = 8.314 \times 2.303 / 7.35\times10^{-5} = 260$ kJ/mol.

3. Creep Mechanisms



At temperature $T > 0.4\ T_m$, thermally activated processes (diffusion) allow dislocations and atoms to overcome obstacles. Dominant mechanism depends on stress and temperature.

- **(i) Dislocation glide** (low T/high σ) : dislocations glide on slip planes, blocked by obstacles – exhaustion of mobile dislocations, hardening – responsible for logarithmic / primary creep at low T (< 0.3 T_m).
- **(ii) Dislocation climb (power-law, dislocation creep)** : edge dislocations stopped by obstacles (precipitates, other dislocations) climb by absorbing/emitting vacancies and then glide on the new plane – rate controlled by **lattice diffusion** ; gives n = 3–8 :

$$\dot{\varepsilon} = A\ (D\ G\ b\ /\ kT)\ (\sigma/G)^n\ ;\ D = D_0\ exp(-Q/RT)$$

Substructure : subgrains with low-angle boundaries (formed by polygonisation), size ≈ 10 G b/σ.

- **(iii) Diffusional creep** (high T, low σ, fine grains) : stress-directed vacancy flow from boundaries under compression to boundaries under tension, so that grains elongate along the stress axis – n = 1 (Newtonian-viscous) :

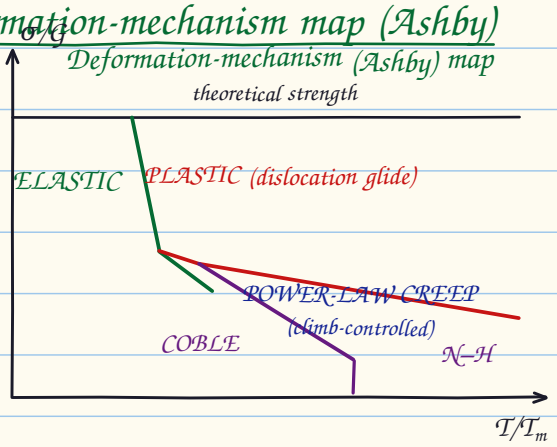
Nabarro–Herring (through lattice) : $\dot{\varepsilon} = A_1\ \Omega\ \sigma\ D_l/(k\ T\ d^2)$ **Coble** (along grain boundaries) : $\dot{\varepsilon} = A_2\ \Omega\ \sigma\ \delta\ D_{gb}/(k\ T\ d^3)$

d = grain size ; Ω = atomic volume ; δ = boundary width. Coble dominates at lower temperatures and finer grains. Coarse grains/ single crystals – diffusional creep decreases ⇒ the reason for directional solidification.

- **(iv) Grain-boundary sliding** : grains slide past each other by viscous shear along boundaries (above ≈ 0.5 T_m, low σ) – must be accommodated by diffusion or slip at triple points; contributes up to 50 % of total strain ; opens wedge cracks and

cavities (Ashby–Verrall model). **Harper–Dorn** creep ($n \approx 1$) at very low stress and $T > 0.9\ T_m$.

Deformation-mechanism map (Ashby)

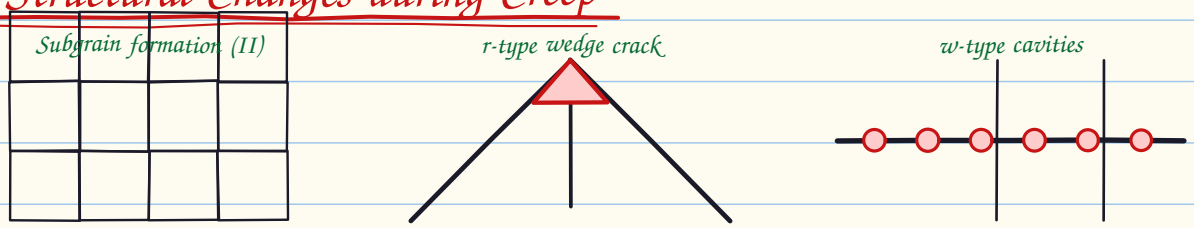


Fields = regions where one mechanism dominates the strain rate for given σ, T, d

Creep : $T > 0.4\ T_m$
Coble (low T, fine d)
N–H (high T, low σ)

- A diagram of normalised stress σ/G versus homologous temperature T/T_m divided into fields where a single mechanism dominates the strain rate (elastic, plastic glide, power-law creep, Nabarro–Herring, Coble) with contours of constant $\dot{\varepsilon}$: lets one read the dominant mechanism and strain rate for given σ, T and grain size d ; used in design of turbine alloys and in geophysics.

4. Structural Changes during Creep



size ∝ G b/σ

triple point, high σ, low T

on ⊥ boundaries, low σ, high T
→ link up → intergranular rupture

- **Primary** : rapid dislocation multiplication, tangles, slip bands ; hardening ⇒ $\dot{\varepsilon}$ falls.
- **Secondary** : recovery by climb and cross-slip ⇒ formation of subgrains (cell structure) with low-angle boundaries – dynamic equilibrium ; subgrain size $\lambda \approx 10\ G\ b/\sigma$ (smaller at high σ) ; slip and grain-boundary migration produce **creep-induced grain-boundary serrations** ; slip bands become coarse and straight at higher T.
- **Tertiary** : necking in ductile metals ; internal cavitation and grain-boundary cracking : **r-type (wedge) cracks** at triple points (high stress, lower temperature, caused by GB sliding) ; **w-type (round) cavities** on boundaries perpendicular to the

stress (low stress, high temperature, vacancy condensation) ; cavities grow by diffusion and link up into intergranular cracks.

- **Microstructural degradation** : coarsening (Ostwald ripening) of precipitates (γ′, carbides), rafting of γ′ in superalloys, dissolution or over-ageing, spheroidisation of pearlite, graphitisation, precipitation of embrittling σ-phase, carburisation/oxidation, recrystallisation of cold-worked metal.
- **Creep fracture** : high stress/low T/short time ⇒ transgranular, ductile (cup-and-cone, 20–40 % RA) ; low stress/high T/long time ⇒ intergranular with little ductility (3–10 %), grain-boundary cavities and wedge cracks. Fracture-mechanism maps (Ashby) show these fields.

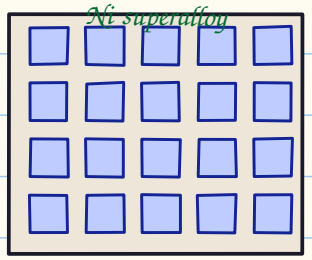
5. Metallurgical Factors Affecting Creep

Factor	Effect	Practice
High melting point	lower diffusivity, high modulus ⇒ better ; creep resistance ∝ T/T_m	Ni, Co, Nb, Mo, W, Ta
Crystal structure	close-packed FCC (low diffusivity) better than BCC (open, fast diffusion) ; low stacking-fault energy hinders climb	austenitic steels, Ni alloys
Grain size	coarse grain better at high T (less GB sliding/diffusion) ; fine grain better below the equicohesive temperature T_E	DS and SX blades
Solid-solution	large atoms (Mo, W, Re) slow diffusion, lower SFE	Ni-Mo-W-Re
Precipitates (γ′, carbides)	coherent ordered particles block climb ; $M_{23}C_6$, M_6C carbides at boundaries stop GB sliding	Ni superalloys, Cr–Mo–V steels
Dispersions (ThO_2, Y_2O_3)	stable to ≈ 0.9 T_m – no coarsening	ODS alloys
Alloying with B, Zr, Hf	segregate to boundaries : ↑ creep ductility, stops cavity growth	superalloy boundaries

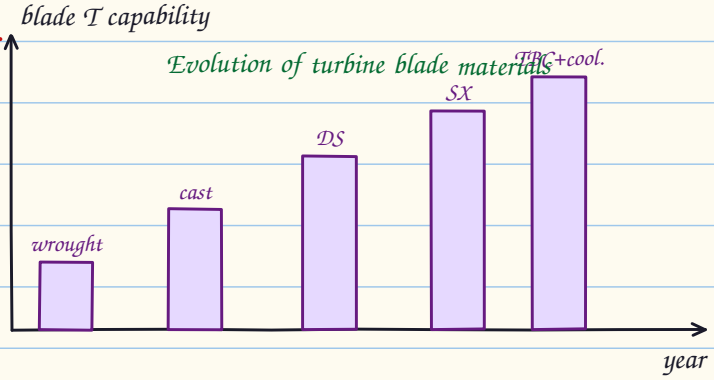
Factor	Effect	Practice
Impurities (S, P, Sn, Pb)	segregate to boundaries ⇒ embrittle	clean melts, vacuum casting
Prior cold work	may raise creep resistance at low T but recovers at high T	not relied on
Heat treatment	solution + double ageing – optimum precipitates	γ' size & distribution
Environment	oxidation, hot corrosion, creep-oxidation cracking	Cr, Al coatings

- **Equicohesive temperature** T_E : temperature where grain boundaries and grain interiors have equal strength ; below T_E ($\approx 0.5\ T_m$) fine grains stronger (Hall–Petch), above it boundaries are the weak path (coarse grains / aligned boundaries better).

6. High-Temperature Alloys



γ' [$Ni_3(Al,Ti)$] cuboids in γ matrix
60–70 % γ', size ≈ 0.3–0.5 µm



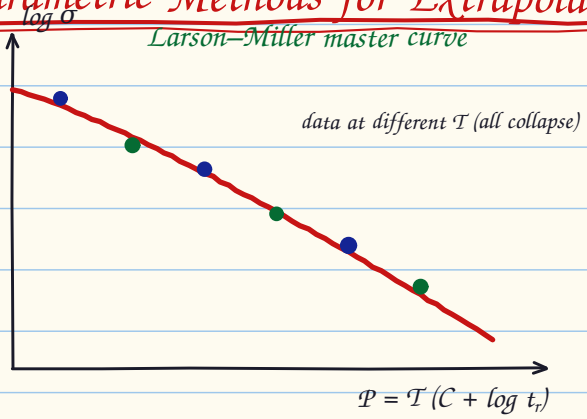
Requirements : high-temperature strength and creep resistance, oxidation / hot-corrosion resistance, microstructural stability, fatigue and thermal-shock resistance, workability / castability, cost.

Class	Examples	Service limit
Low-alloy ferritic steels	1–1.25Cr–0.5Mo, 2.25Cr–1Mo (boilers, pipes)	≈ 565 °C
Advanced 9–12Cr steels	9Cr–1Mo–V (P91), P92	≈ 600–620 °C
Austenitic stainless	304H, 316, 321, 347, 310	≈ 650–700 °C
Iron-base superalloys	A-286, Incoloy 800	≈ 700 °C

Class	Examples	Service limit
Nickel-base superalloys	Inconel 718, 625, Waspaloy, Rene 41/80, IN-738, MAR-M247, CMSX-4	up to 1100 °C (0.85 T_m)
Cobalt-base	Haynes 25 (L-605), Stellite, X-40	vanes, hot corrosion ≈ 1000 °C
Refractory metals	Nb, Mo, Ta, W alloys (C-103, TZM)	> 1200 °C (oxidation protection needed)
Intermetallics / ceramics	TiAl, Ni_3Al, $MoSi_2$; SiC, Si_3N_4, CMC (SiC/SiC)	up to 1300–1500 °C

- **Ni-base superalloy microstructure** : FCC γ matrix (Ni–Cr–Co solid solution with Mo, W, Re) + coherent ordered γ′ $Ni_3(Al,Ti)$ cuboids (60–70 vol %) that strengthen up to 0.8 T_m (strength rises with T – anomalous yield) ; carbides (MC, $M_{23}C_6$) on grain boundaries ; B, Zr add boundary cohesion ; TCP phases (σ, μ) avoided.
- **Processing evolution** : wrought → conventional cast → **directionally solidified (DS)** columnar grains parallel to stress (no transverse boundaries) → **single crystal (SX)** (no boundaries ⇒ no B, C, Zr needed – CMSX-4 with Re) ; each step ≈ +20–30 °C ; **ODS** alloys (MA6000) ; powder-metallurgy discs (René 95) ; film-cooled blades, thermal barrier coatings (YSZ over MCrAlY bond-coat) add 100–150 °C ; **aluminide / MCrAlY coatings** for oxidation.
- Selection : steam pipes (P91), gas turbine blades (SX Ni-base + TBC), discs (IN-718/ René 95), combustors (Hastelloy X, Haynes 230), exhaust/automotive turbo (Inconel 713), heating elements (Ni–Cr 80/20 Nichrome, Kanthal Fe–Cr–Al), rocket nozzles (Nb, C/C).

7. Parametric Methods for Extrapolating Creep Data



$P = T\,(C + \log t_r)$

T in K, t_r in hours, $C \approx 20$

Sherby–Dorn : $P_{SD} = \log t_r - 0.434Q/RT$

Manson–Haferd : $(\log t_r - \log t_a)/(T - T_a)$

Monkman–Grant : $\dot{\varepsilon}_s \cdot t_r = C_{MG}$

Long-life data (10^5 h) cannot be obtained by direct testing ⇒ short-time high-temperature tests are extrapolated using **time–temperature parameters** :

Larson–Miller : $P = T\,(C + \log t_r)$ (T in K, t_r in h, $C \approx 20$)

- Principle : $\log t_r + \text{const} = Q/(2.3RT)$ – same rupture life at different (T, t) combinations at the same stress ⇒ a single **master curve** $\log \sigma$ vs P ; read P at the design stress and compute the life at service T. Others : Sherby–Dorn, Manson–Haferd, Orr–Sherby–Dorn.

Ex 3. A stress gives t_r = 1000 h at 900 K : $P = 900(20 + 3) = 20\,700$. At 850 K : $\log t_r = 20\,700/850 - 20 = 4.35 \Rightarrow t_r = 2.25\times10^4$ h. To last 25 000 h ($\log$ = 4.40) : $T = 20\,700/24.4 = 848$ K = 575 °C.

Ex 4. Steel : at P = 18 000 (T in K) allowable stress read from the master curve ; for T = 700 K life $\log t_r = 18\,000/700 - 20 = 5.71 \Rightarrow 5.1\times10^5$ h.

8. Stress-Rupture Testing

- **Stress-rupture (creep-rupture) test** : specimens loaded at constant load/temperature until fracture ; records time to rupture t_r, elongation and reduction of area at fracture, sometimes minimum creep rate. Higher stresses than creep tests, shorter times (up to 10^3–10^5 h).
- Result presented as **$\log \sigma$ vs $\log t_r$** at several temperatures – straight lines (parallel) ; a break in slope at long times indicates a change in mechanism : over-ageing, recrystallisation, oxidation, transition from transgranular to intergranular failure ; notched specimens test **notch sensitivity** (notch strengthening vs weakening). The **rupture strength** (e.g. stress for rupture in 1000

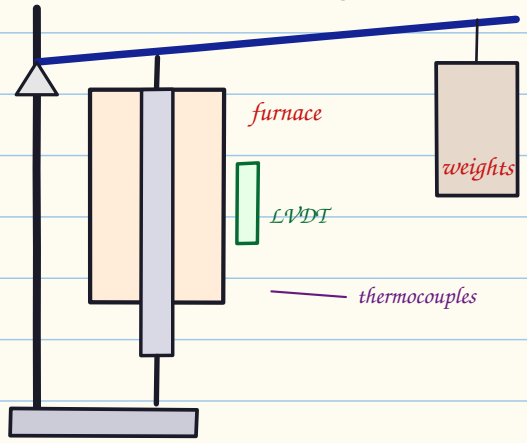
/ 10 000 / 100 000 h at T) is the design property ; also stress for 1 % creep in 10^4 h.

- **Procedure** (ASTM E139, E292, ISO 204) : round specimen (gauge dia 6.35 mm, gauge length 25–50 mm, sometimes notched), thermocouples attached, furnace set ±2–3 °C (to 1000 °C ; ±5 up to 1200), allowed to equilibrate, load applied smoothly via lever/weights, elongation measured, timer stops at rupture (microswitch), fractography and metallography afterwards.
- **Design stresses** : allowable stress = lowest of (σ_y/1.5, 100 % average stress for 10^5 h rupture ÷ 1.5 (or 0.67 of mean), stress for 1 % creep in 10^5 h). (ASME code.)

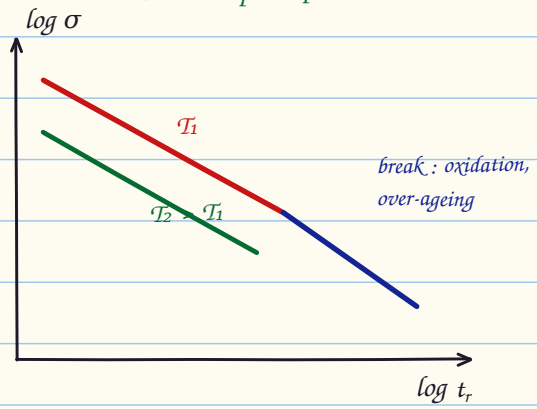
Ex 5. A steel steam pipe designed for 10^5 h at 550 °C (P from master curve) rupture strength 90 MPa; allowable design stress = 0.67 × 90 = 60 MPa (or the stress for 1 % creep in 10^5 h, whichever is lower).

9. Creep Testing Machines

Lever-arm creep testing machine



Stress–rupture plot



Components of a creep testing machine

- **Loading system** : dead-weight lever-arm (lever ratio 3 : 1 to 20 : 1 ; load pan with weights) ; spring/hydraulic or screw systems (for stress-relaxation and constant-rate tests) ; servo-hydraulic frames ; for **constant true stress** a cam, lever-contour (Andrade–Chalmers) or electronic controller adjusts the load as the area changes.
- **Furnace** : split tubular resistance furnace (Kanthal / Pt / Mo wire) ; multiple zones ; PID temperature controller ; temperature uniform within ±2 °C over the gauge length at ≤ 1000 °C ; thermocouples (type K/N/S) on the specimen

shoulders ; inert or vacuum atmosphere for refractory metals.

- **Extensometry** : specimen shoulders carry arms outside the furnace ⇒ *LVDT / capacitive / optical* extensometer, strain sensitivity 10^{-6}–10^{-5} ; data logger plotting ε–t. Alignment grips / universal joints to avoid bending ; pull-rods of heat-resistant alloy.
- **Test types** : **creep test** (ε–t, long term 1000–10^5 h, small stress to get minimum creep rate and 0.1–1 % strains), **stress-rupture test, stress-relaxation test** (strain fixed, stress decays – bolts), **step-load / step-temperature tests** for n and Q, **constant strain-rate, creep-fatigue,** small-punch and impression creep for in-service components, miniature specimens.
- **Precautions** : constant temperature, accurate axial alignment, no shock loading, correction for thermal expansion and drift ; multiple specimens in line (multi-station racks) ; tests run 24 h/day. Accelerated testing : raise stress and/or T and use parameters (LMP).
- **Short-time tension tests at high temperature** – determine hot yield and tensile strength, ductility ; hot hardness test.

10. Additional Solved Problems

P1. The steady creep rate of a metal is $2\times10^{-7}\ h^{-1}$ at 100 MPa and 600 °C ; n = 5. Find the rate at 150 MPa : $(1.5)^5 = 7.59$ ⇒ $1.5\times10^{-6}\ h^{-1}$.

P2. Q = 250 kJ/mol ; rate at 800 K is $10^{-8}\ h^{-1}$. At 850 K : ratio = exp[(250 000/8.314)(1/800 − 1/850)] = exp(30 070 × 7.35×10^{-5}) = exp(2.21) = 9.1 ⇒ $9.1\times10^{-8}\ h^{-1}$.

P3. A bar length 500 mm creeps at $5\times10^{-8}\ s^{-1}$ for 1000 h : strain = $5\times10^{-8}\times3.6\times10^{6}$ = 0.18 (18 %) – too much ; a rate of $10^{-10}\ s^{-1}$ ⇒ 3.6×10^{-4} (0.036 %) in 1000 h.

P4. Monkman–Grant : $\dot{\varepsilon}_s = 10^{-6}\ h^{-1}$, $C_{MG} = 0.05$ ⇒ $t_r = 0.05/10^{-6} = 5\times10^{4}$ h.

P5. LMP (C = 20) : stress rupture at 700 °C (973 K) for 1000 h : P = 973 × 23 = 22 380. Life at 650 °C (923 K) = $10^{22\,380/923\ -\ 20} = 10^{4.25} = 1.8\times10^{4}$ h.

Short answers – Quick Revision

- *Why does creep occur only at high temperature?* Needs thermally activated diffusion (climb, vacancy flow, GB sliding) – significant above 0.4 T_m.

- *Why is stage II called steady state?* Work hardening equals recovery ⇒ constant dislocation structure and rate.
- *Why are single crystals used for turbine blades?* No grain boundaries ⇒ no GB sliding, cavitation or diffusional creep.
- *Difference between creep and stress relaxation?* Creep – strain increases at constant stress ; relaxation – stress decays at constant strain.
- *What does the LMP do?* Relates rupture life, stress and temperature in one master curve for extrapolation.

11. Quick Formula Sheet – Unit V

Item	Formula
Homologous temperature	$T/T_m > 0.4 \Rightarrow$ creep
Steady creep rate	$\dot{\varepsilon}_s = A\sigma^n \exp(-Q/RT)$
Garofalo	$\varepsilon = \varepsilon_0 + \varepsilon_t(1 - e^{-rt}) + \dot{\varepsilon}_s t$
Nabarro–Herring	$\dot{\varepsilon} \propto \sigma D_l / d^2$
Coble	$\dot{\varepsilon} \propto \sigma D_{gb} / d^3$
Monkman–Grant	$\dot{\varepsilon}_s t_r = C$
Larson–Miller	$P = T(C + \log t_r)$, $C \approx 20$
Activation energy	$Q = R\,\ln(\dot{\varepsilon}_2/\dot{\varepsilon}_1)/(1/T_1 - 1/T_2)$

12. Important Questions

- Define creep. Draw a typical creep curve and explain the three stages with the underlying mechanisms.
- Discuss the effect of stress and temperature on the creep curve ; Norton law and activation energy for creep.
- Explain the creep mechanisms : dislocation glide/climb, diffusional (Nabarro–Herring, Coble), grain-boundary sliding ; deformation-mechanism maps.
- Describe the structural changes during creep ; creep fracture (wedge cracks, cavities).
- Discuss the metallurgical factors affecting creep resistance ; why single crystals / DS blades?

- Write notes on high-temperature alloys (steels, Ni-base superalloys) and their strengthening.
- Explain the stress-rupture test and Larson–Miller parameter. Describe a creep testing machine with a neat sketch.